## Example 1

The officers of a high school senior class are planning to rent buses and vans for a class trip. Each bus can transport 45 students, requires 4 supervisors and costs Birr 4,000 to rent. Each van can transport 12 students, requires 1 supervisor, and costs Birr 600 to rent. The officers must plan to accommodate at least 420 students. Since only 48 parents have volunteered to serve as supervisors, the officers must plan to use at most 48 supervisors. How many vehicles of each type should the officers rent in order to minimize the transportation costs? What is the minimum transportation cost?

### Solution

To solve this application problem first we define the decision variables as $x$ is the number of buses to rent, $y$ is the number of vans to rent.

The objective function is the cost function, $Z = 4000x + 600y$

These are two limitations, number of students to be accommodated is at least 420 and supervisors are at most 48, and this is expressed as

$4x + y \leq 48$, Supervisor's limitation

$45x + 12y \geq 420$, Student's accommodation

$x \geq 0, y \geq 0$. Non negativity

The LPP is

Minimize $Z = 4000x + 600y$ (1)

Subject to $\begin{cases} y \leq -4x + 48 \\ y \geq \dfrac{-15}{4}x + 35 \\ x \geq 0, y \geq 0 \end{cases}$

The feasible region is determined by drawing the lines $y = -4x + 48$ and $y = -\frac{15}{4}x + 35$. By shading the regions below the first line and above the second we get the Figure 4.21.

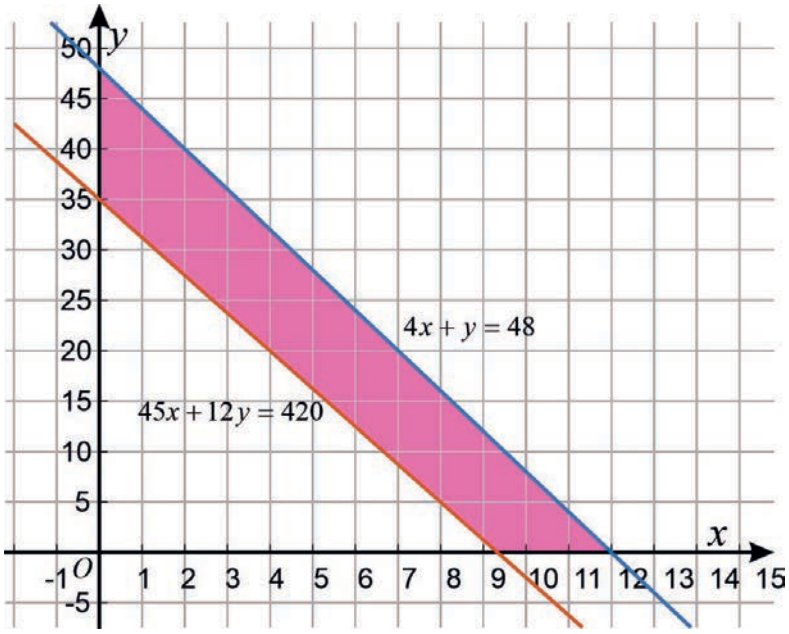



Figure 4.21

The proposed optimal points are (0, 48), (0, 35), (12, 0) and $(\frac{28}{3}, 0)$. The values of the objective function at these points are:

At (0, 48), $Z = 600 \cdot 48 = 28{,}800$.

At (0, 35), $Z = 600 \cdot 35 = 21{,}000$.

At (12, 0), $Z = 4000 \cdot 12 = 48{,}000$.

At $(\frac{28}{3}, 0)$, $Z = 4000 \cdot \frac{28}{3} = 37{,}333.33 \ldots \approx 37{,}333$. Thus, the minimum cost is 21,000 Birr.

## Example 2

There are two factories one is located at industrial Park 1 and the other at industrial Park 2. From these parks, a certain commodity is to be delivered to each of the three markets situated at A, B and C. The weekly requirements of the markets are respectively 5, 5 and 4 units of the commodity while the production capacity of the factories at Park 1 and Park 2 are respectively 8 and 6 units. The cost of transportation per unit is given in the table below:

| From/To | Cost in Birr | | |
|---|---|---|---|
| | A | B | C |
| Park 1 | 160 | 100 | 150 |
| Park 2 | 100 | 120 | 100 |

How many units should be transported from each factory to each market in order that the transportation cost is minimum? What will be the minimum transportation cost?

## Solution

The problem can be explained diagrammatically in Figure 4.22.

Let $x$ units and $y$ units of the commodity be transported from the factory at Park 1 to the markets at A and B respectively. Then $(8 - x - y)$ units will be transported to market at C. From these we have $x \geq 0, y \geq 0$, and $8 - x - y \geq 0$ equivalent to $x + y \leq 8$.

Now, the weekly requirement of the market at A is 5 units of the commodity. Since $x$ units are transported from the factory at Park 1, the remaining $(5 - x)$ units need to be transported from the factory at Park 2. Obviously, $5 - x \geq 0, i.e. x \leq 5$.

Similarly, $(5 - y)$ units are to be transported from the Park 2 to the market at B and $6 - (5 - x + 5 - y) = x + y - 4$ units are to be transported from the Park 2 to the market at C. Thus, $5 - y \geq 0, x + y - 4 \geq 0, i.e. y \leq 5, x + y \geq 4$.

Total transportation cost $Z$ is given by

$$Z = 160x + 100y + 150(8 - x - y) + 100(5 - x) + 120(5 - y) + 100(x + y - 4)$$

$$= 10x - 70y + 1900$$

$$= 10(x - 7y + 190)$$

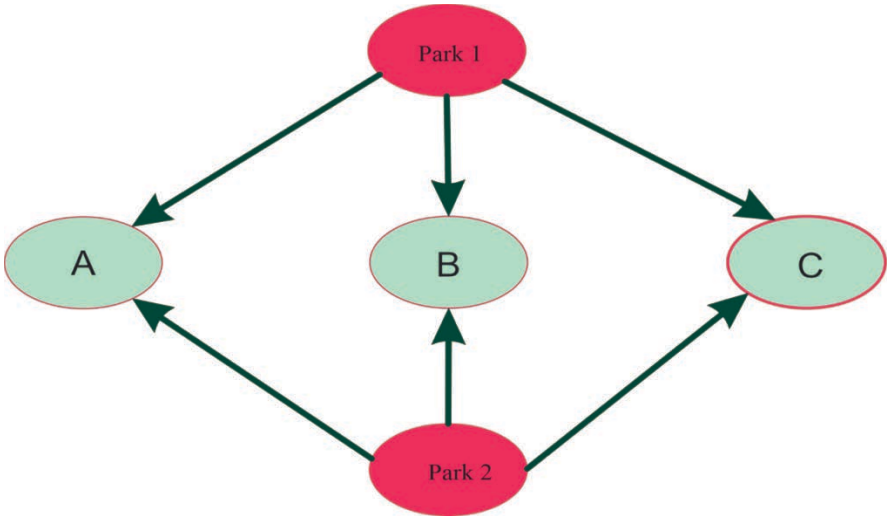



Figure 4.22

Therefore, the problem is expressed as: Minimize $Z = 10(x - 7y + 190)$.

Subject to the constraints: $\begin{cases} x + y \leq 8 \\ x \leq 5,\ y \leq 5 \\ x + y \geq 4 \\ x \geq 0, y \geq 0 \end{cases}$

Drawing the constraint inequality in the $xy$ –plane you obtain the feasible region enclosed by *ABCDEF*. Observe that the feasible region is bounded. The coordinates of the corner points of the feasible region are (0, 4), (0, 5), (3, 5), (5, 3), (5, 0) and (4, 0).

Let us evaluate $Z$ at these points.

| Corner points | $Z = 10x - 70y + 1900$ |
|---|---|
| (0, 4) | 1620 |
| (0, 5) | 1550 |
| (3, 5) | 1580 |
| (5, 3) | 1740 |
| (5, 0) | 1950 |
| (4, 0) | 1940 |

From the table, we see that the minimum value of $Z$ is 1550 at the point (0, 5). Hence, the optimal transportation strategy will be to deliver 0, 5 and 3 units from the factory at Park 1 and 5, 0 and 1 units from the factory at Park 2 to the market at $A$, $B$ and $C$ respectively. Corresponding to this strategy, the transportation cost would be minimum, i.e., Birr 1550.

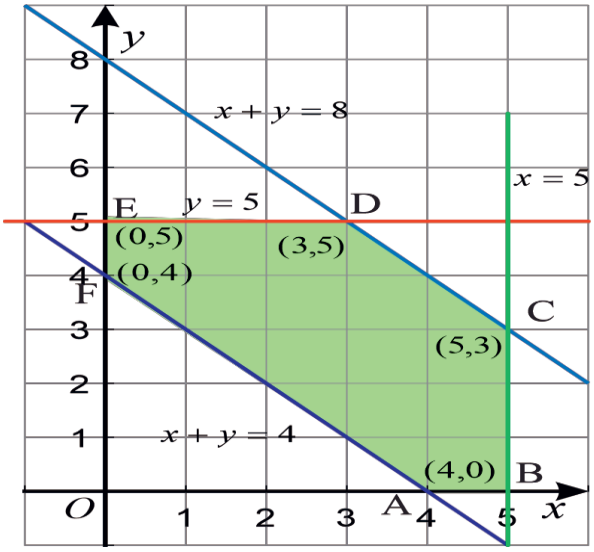



Figure 4.23

## Exercise 4.12

1. An oil company has two depots A and B with capacities of 7000L and 4000L respectively. The company is to supply oil to three petrol stations, D, E and F whose requirements are 4500L, 3000L and 3500L respectively in Addis Ababa. The distances (in km) between the depots and the petrol stations are given in the following table. Assuming that the transportation cost of 10 liters of oil is Birr 2 per km, how should the delivery be scheduled in order that the transportation cost is minimum? What is the minimum cost?

| Distance in kilometer | | |
|---|---|---|
| From/To | A | B |
| D | 7 | 3 |
| E | 6 | 4 |
| F | 3 | 2 |

## 4.3.2 Solving Linear Programming Problems Using Spreadsheet

In the previous sections, you discussed how to solve LPP graphically. Now, you will see how to solve LPP by the help of computer applications in particular spreadsheet.

**Activity 4.6**

Discuss the main features of Microsoft Excel in a group.

A mathematical model implemented in a spreadsheet is called a spreadsheet model. Major spreadsheet packages come with a built-in optimization tool called Solver. Now we demonstrate how to use Excel spreadsheet modeling and Solver to find the optimal solution of optimization problems.

If the model has two variables, the graphical method can be used to solve the problem. Very few real world problems involve only two variables. For problems with more than two variables, we need to use complex techniques and tedious calculations to find the optimal solution. The spreadsheet solver approach makes solving optimization problems a fairly simple task.

Nowadays, there are various computer applications available for solving LP problems. MS Excel useful to solve LP problems with its "SOLVER" application. Hence, the Solver application should be installed and appear in the Excel toolbar.

To install solver in Excel, click **"File"** and select **"Excel Options",** where click on **"Add-Ins".** You can find ***"Manage: Excel Add-Ins"*** at the bottom of the browser. Then, click on **"Go",** where you find an Add-Ins message box and select the option ***"Solver Add-In"*** and finally click ***"OK".*** Now, you can find the icon ***"Solver"*** in ***"Analysis"*** the subsection of Data toolbar in the main menu to solve your LP problems.

The first step is to organize the spreadsheet to represent the model. We use separate cells to represent decision variables, create a formula in a cell to represent the objective function and create a formula in a cell for each constraint left hand side.

Once the model is implemented in a spreadsheet, next step is to use the **Solver** to find the solution. In the Solver, we need to identify the locations (cells) of objective function, decision variables, nature of the objective function (maximize/minimize) and constraints.

## Example 1

A Textile PLC produces bedsheets of types *S* and *L*. The wholesale price is Birr 360 per bedsheet for *S* and Birr 800 for *L*. Two time constraints result from the use of two machines $M_1$ and $M_2$. On $M_1$ needs 2 min for an *S* bedsheet and 8 min for an *L* bedsheet. On $M_2$ needs 5 min for an *S* bedsheet and 2 min for an *L* bedsheet. Determine production figures $x$ and $y$ for *S* and *L*, respectively; i.e., number of bedsheets produced per hour, so that the hourly revenue is maximized.

### Solving LP problem in excel

The objective function and the constraint for the production problem is

Maximize $Z = 360x + 800y$

$2x + 8y \leq 60$ on machine $M_1$,

$5x + 2y \leq 60$ on machine $M_2$,

$x \geq 0, y \geq 0.$

**Step 1:** Put the problem in Excel. Put the objective function coefficients into a row with **at least 2 blank rows above it** with the constraint coefficients below. Label the rows down the left hand side in column 1. Leave one blank column after the last variable and label it **sum.**

Then put in the right hand side (RHS). Put names for each variable above the variables in the row just above the objective function coefficients. Label that row **Names of decision variables**. The resultant spreadsheet is

| Step 1 | | | | | |
|---|---|---|---|---|---|
| | | | | | |
| Name of Decision Variable | x | y | Sum | RHS | |
| Objective to maximize | 360 | 800 | | | |
| Available time on M1 | 2 | 8 | | 60 | |
| Available time on M2 | 5 | 2 | | 60 | |
| | | | | | |

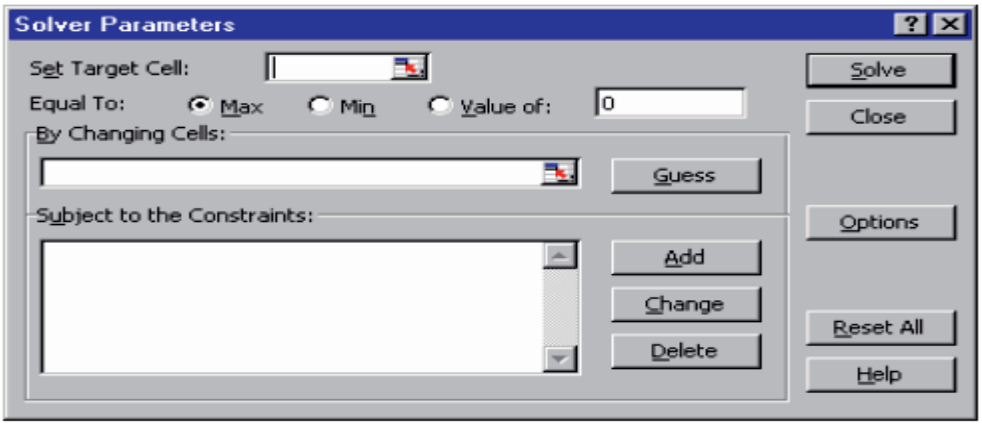
**Step 2:** Now, label the row just above tableau **Variable values to manipulate**. **Enter 0 values above the variables**. These are the cells that Excel will "change" to find the optimum solution to the problem.

| Variable values to manipulate | 0 | 0 | | |
|---|---|---|---|---|
| Name of Decision Variable | x | y | Sum | RHS |
| Objective to maximize | 360 | 800 | | |
| Available time on M1 | 2 | 8 | | 60 |
| Available time on M2 | 5 | 2 | | 60 |

**Step 3:** Now, construct Excel cell entries to add up each LP model equations. Place these in the column named Sum. These will involve adding the numbers in each equation times the numbers from the Variable values to manipulate row. Namely, with Variable values in row 8 and the objective function in row 10 enter the equation.

| Step 1-3 | | | | |
|---|---|---|---|---|
| Variable values to manipulate | 0 | 0 | | |
| Name of Decision Variable | x | y | Sum | RHS |
| Objective to maximize | 360 | 800 | 0 | |
| Available time on M1 | 2 | 8 | 0 | 60 |
| Available time on M2 | 5 | 2 | 0 | 60 |
| | | | | |

**Step 4:** Activate the solver. To do this, go to Tools in the toolbar and click on Solver. The following dialogue then appears:



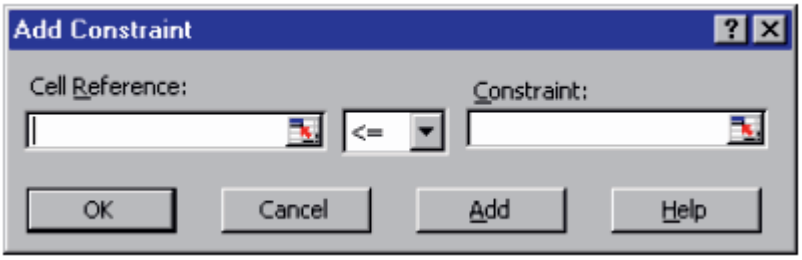

**Step 5:** Define where the objective function is by defining the variable dialogue box called **Set Target Cell** as the cell number where you added the objective function up.

**Step 6:** Choose whether to maximize or minimize using the buttons just below the Set Target Cell box.

**Step 7:** Identify the decision variables by entering the range in which they fall in the **By Changing Cells box.**

**Step 8:** Enter consideration of the constraint equations into the model by clicking the Add

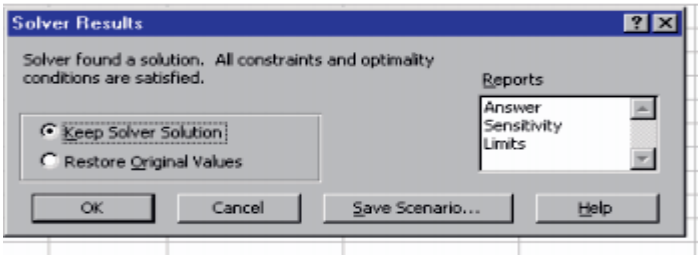
button to the right of the Subject to the Constraints box. You then get the dialogue below;



This step is multistep that depends on the number of constraints.

**Step 9:** Review the problem and when all looks right, solve by clicking the solve button.

**Step 10:** Excel will solve LP problem based on the formulas you inputted. When Excel finds an optimal solution, the following appears.




**Step 11:** Choose desired output reports. Highlight both the Answer Report and Sensitivity Report. Click on Keep Solver Solution and OK then the Reports will be generated. You will see that Excel has entered optimal **Decision Levels** and Total Resource Use in proper cells. And added two new sheets one with the answer and the other for sensitivity. For our example we have get the following two results,

**Table1. Result in proper sheet**

| | | | | |
|---|---|---|---|---|
| Variable values to manipulate | 10 | 5 | | |
| Name of Decision Variable | x | y | Sum | RHS |
| Objective to maximize | 360 | 800 | 7600 | |
| Available time on M1 | 2 | 8 | 60 | 60 |
| Available time on M2 | 5 | 2 | 60 | 60 |

**Table 2**

**Microsoft Excel 15.0 Sensitivity Report**

**Worksheet: [LPP Example 1.xlsm]Sheet1**

**Report Created: 8/8/2021 2:40:01 PM**

Variable Cells

| Cell | Name | Final Value | Reduced Gradient |
|---|---|---|---|
| $B$8 | Variable values to manipulate x= | 10 | 0 |
| $C$8 | Variable values to manipulate y= | 5 | 0 |

Constraints

| Cell | Name | Final Value | Lagrange Multiplier |
|---|---|---|---|
| $D$11 | Available time on M1 Sum | 60 | 91.11111111 |
| $D$12 | Available time on M2 Sum | 60 | 35.55555556 |

**Microsoft Excel 15.0 Answer Report**

**Worksheet: [LPP Example 1.xlsm]Sheet1**

**Report Created: 8/8/2021 2:40:01 PM**

**Result: Solver found a solution. All Constraints and optimality conditions are satisfied.**

**Solver Engine**

Engine: GRG Nonlinear

Solution Time: 0.047 Seconds.

Iterations: 0 Sub-problems: 0

**Solver Options**

Max Time Unlimited, Iterations Unlimited, Precision 0.000001

Convergence 0.0001, Population Size 100, Random Seed 0, Derivatives Central

Max Sub-problems Unlimited, Max Integer Sols Unlimited, Integer Tolerance 1%, Assume Non-Negative

Objective Cell (Max)

| Cell | Name | Original Value | Final Value |
|---|---|---|---|
| $D$10 | Objective to maximize Sum | 7600 | 7600 |

Variable Cells

| Cell | Name | Original Value | Final Value | Integer |
|---|---|---|---|---|
| $B$8 | Variable values to manipulate x= | 10 | 10 | Contin |
| $C$8 | Variable values to manipulate y= | 5 | 5 | Contin |

Constraints

| Cell | Name | Cell Value | Formula | Status | Slack |
|---|---|---|---|---|---|
| $D$11 | Available time on M1 Sum | 60 | $D$11<=$E$11 | Binding | 0 |

| | | | | | |
|---|---|---|---|---|---|
| | Available time on | | | | |
| $D$12 | M2 Sum | 60 | $D$12<=$E$12 | Binding | 0 |

## Exercise 4.13

Solve the Linear Programing Problem by using Excel solver or spreadsheet.

a. Maximize $z = 4x_1 + 5x_2$

Subject to

$2x_1 + x_2 \leq 0,$
$x_1 \leq 4,$
$x_1 \geq 0,\ x_2 \geq 0.$

b. Maximize $z = 5x_1 + 6x_2$

Subject to

$3x_1 + 4x_2 \leq 18,$
$2x_1 + x_2 \leq 7\ ,$
$x_1 \geq 0\ ,\ x_2 \geq 0.$

c. Minimize $z = 2x_1 + 3x_2$

Subject to

$4x_1 + 2x_2 \geq 12,$

$x_1 + 4x_2 \geq 6\ ,$

$x_1 \geq 0,\ x_2 \geq 0.$

## Summary

- Half-Plane: The region on a side of a line in the $xy$-plane.
- A system of linear inequalities is a collection of two or more linear inequalities to be solved simultaneously.
- A graphical solution of a system of linear inequalities is the graph of all ordered pairs $(x, y)$ that satisfy all the inequalities.
- A point of intersection of two or more boundary lines of a solution region is called a vertex (or a corner point) of the region.
- A number $M = f(c)$ for some $c$ in $I = \{x : a \leq x \leq b\}$ is called the maximum value of $f$ on $I$, if $M \geq f(x)$, for all $x$ in $I$.
- A number $m = f(d)$ for some $d$ in $I = \{x : a \leq x \leq b\}$ is called the minimum value of $f$ on $I$, if $m \leq f(x)$, for all $x$ in $I$
- A value which is either a maximum or a minimum is called an optimum (or extremum) value.
- A linear programming problem is one that is concerned with finding the optimal value (maximum or minimum) of a linear function of several variables (called objective function) subject to the conditions that the variables are non-negative and satisfy a set of linear inequalities (called linear constraints).
- Variables are sometimes called decision variables and are non-negative. A few important linear programming problems are:
  - i) Diet problems
  - ii) Manufacturing problems
  - iii) Transportation problems
  - iv) Allocation problems
- The common region determined by all the constraints including the non-negative constraints $x \geq 0, y \geq 0$ of a linear programming problem is called the feasible region (or solution region) for the problem.
- Any point outside the feasible region is an infeasible solution.

- Any point in the feasible region that gives the optimal value (maximum or minimum) of the objective function is called an optimal solution.

## Review Exercise

1. Determine whether each ordered pair is a solution to the inequality $y > x - 1$.
   a. $(0, 1)$ b. $(-4, -1)$ c. $(-2, 0)$
   d. $(4,2)$ e. $(3, 0)$
2. Graph each of the following linear inequalities.
   a. $y \leq x - 1$ b. $y < \frac{3}{5}x + 2$
   c. $y \leq -\frac{1}{2}x + 4$ d. $y \geq -\frac{1}{3}x - 2$
3. Solve each system by graphing.
   i. $\begin{cases} -3x + 5y > 10 \\ x > -1 \end{cases}$ ii. $\begin{cases} 2x + 4y > 4 \\ y \leq -\frac{1}{2}x - 2 \end{cases}$
   iii. $\begin{cases} -2x + 6y < 0 \\ 6y > 2x + 4 \end{cases}$ iv. $\begin{cases} 2x + y > -6 \\ -x + 2y \geq -4 \end{cases}$
4. Solve the inequality $\begin{cases} y \leq -5x + 2, \\ x \geq 0, \\ y \geq 1. \end{cases}$
5. Discuss steps of solving linear programming problems.
6. Sultan works two part time jobs. One is at a gas station that pays Birr 11 an hour and the other is Computer technician for Birr 16.5 an hour. Between the two jobs, Sultan wants to earn at least Birr 3300.00 in four weeks. How many hours does Sultan need to work each job to earn at least Birr 3300. Assume Sultan has 15 working hours per day and works 5 days per week. Write an inequality which models the situation. Graph the inequality and list three solutions to the inequality.
7. Fikadu was told by his Doctor that he needs to exercise enough to burn 500 calories each day. Thus, he prefers either to run or bike and burns 15 calories per

minute while running and 10 calories a minute while biking. He can allocate at most 40 minutes per day for exercises. Find the inequality that models the situations. Graph the inequality and list three solutions to the inequality.

8. Find the maximum and minimum values of the objective functions subject to the given constraints.

i) Objective functions $z = 6x + 4y$

$$\text{Subject to } \begin{cases} x + y \le 5 \\ y \ge 2 \\ x \ge 0, y \ge 0 \end{cases}$$

ii) Objective functions $z = 3x + 5y$

$$\text{Subject to } \begin{cases} x - 2y \le 6 \\ x \le 10 \\ y \ge 1 \\ x \ge 0, y \ge 0 \end{cases}$$

iii) Objective function

$$z = 6x + 4y$$

$$\text{Subject to: } \begin{cases} 2x + 3y \le 30 \\ 3x + 2y \le 24 \\ x + y \ge 3 \\ x \ge 0, y \ge 0 \end{cases}$$

iv) Objective function

$$z = 3x + 9y$$

$$\text{Subject to: } \begin{cases} x + 4y \le 8 \\ x + 2y \le 4 \\ x \ge 0, y \ge 0 \end{cases}$$

9. Maximize or Minimize, $Z = 3x + 4y$,

$$\text{Subject to: } \begin{cases} x + 2y \le 14 \\ 3x - y \ge 0 \\ x - y \le 2 \\ x \ge 0, y \ge 0 \end{cases}$$

10. Minimize and Maximize $Z = 5x + 10y$,

$$\text{Subject to: } \begin{cases} x + 2y \le 120 \\ x + y \ge 60 \\ x - 2y \le 0 \\ x \ge 0, y \ge 0 \end{cases}$$

11. Minimize and Maximize $Z = x + 2y$,

$$\text{Subject to: } \begin{cases} x + 2y \le 100 \\ 2x - y \le 0 \\ 2x + y \le 200 \\ x \ge 0, y \ge 0 \end{cases}$$

12. Maximize, $Z = -x + 2y$,

Subject to the constraints: $\begin{cases} x \geq 3 \\ x + y \geq 5 \\ x + 2y \leq 6 \\ y \geq 0 \end{cases}$

13. Maximize $Z = x + y$,

Subject to $\begin{cases} x - y \leq -1 \\ -x + y \leq 0 \\ x \geq 0, y \geq 0 \end{cases}$

14. Find the maximum and minimum values of

a. Objective function $Z = 6x + 10y$

Subject to $\begin{cases} 2x + 5y \leq 10 \\ x \geq 0,\ y \geq 0 \end{cases}$

b. Objective function $Z = 4x + y$

Subject to: $\begin{cases} x - y \leq 40 \\ 2x + 3y \leq 72 \\ x \geq 0,\ y \geq 0 \end{cases}$

15. A health fitness center wants to produce a guide to healthy living. The center intends to produce the guide in two formats: a short video and a printed book. The center needs to decide how many of each format to produce for sale. Estimates show that no more than 10,000 copies of both items together will be sold. At least 4,000 copies of the video and at least 2,000 copies of the book could be sold, although sales of the book are not expected to exceed 4000 copies. Let $x$ be the number of videos sold, and $y$ the number of printed books sold.

a. Write down the constraint inequalities that can be deduced from the given information.

b. Represent these inequalities graphically and indicate the feasible region clearly.

c. The health fitness center is seeking to maximize the income, $I$, earned from the sales of the two products. Each video will be sold for Birr 50 and each book for Birr 30. Write down the objective function for the income.

d. What maximum income will be generated by the two guides?

e. Solve the problem by using spreadsheets solver.

16. A housing corporation plans to build a maximum of 11 new houses in a large city. This company will build these houses in one of three sizes for each location – a convenience house, standard house, and luxury house. The convenience house requires Birr 4.125 million to build and 30 employees to operate. The standard house requires Birr 8.25 million to build and 15 employees to operate. The luxury house requires Birr 12.375 million to build and 45 employees to operate. The corporation can dedicate Birr 82.5 million in construction capital, and 300 employees to staff the stores. On the average, the convenience house nets Birr 1.2 million annually, the standard nets Birr 2 million annually, and the luxury nets Birr 2.6 million annually. How many of each should they build to maximize revenue?( Challenging, extend your excel knowledge to three variables)

17. To meet the requirements of a specialized diet a meal is prepared by mixing two types of cereal, Wheat and Sorghum (Mashila). The mixture must contain $x$ packets of Wheat cereal and $y$ packets of Sorghum cereal. The meal requires at least 15 g of protein and at least 72 g of carbohydrates. Each packet of Wheat cereal contains 4 g of protein and 16 g of carbohydrates. Each packet of Sorghum cereal contains 3 g of protein and 24 g of carbohydrates. There are at most 5 packets of cereal available.

    i) Write down the constraint inequalities.

    ii) Identify the feasible region.

    iii) If Wheat cereal costs Birr 40 per packet and Sorghum cereal also costs Birr 40 per packet, use the graph to determine how many packets of each cereal must be used so that the total cost for the mixture is a minimum.

    iv) Use the graph to determine how many packets of each cereal must be used so that the total cost for the mixture is a maximum (give all possibilities).

# UNIT 5

# MATHEMATICAL APPLICATIONS IN BUSINESS

Financial Center of Ethiopia and the new Head Quarter of CBE

## Unit Outcomes

**By the end of this unit, you will be able to:**

- Know common terms related to business.
- Know basic concepts in business.
- Understand time value of money.
- Apply mathematical principles and theories to practical situations.

## Unit Contents


**5.1** Basic Mathematical Concepts in Business

**5.2** Time Value of Money

**5.3** Saving, Investing and Borrowing Money

**5.4** Taxation

Summary

Review Exercise

- Amortization
- Annual interest rate
- Appreciation
- Checking accounts
- Compound proportion
- Compound interest
- Constant of variation
- Continued ratio
- Creditor
- Debt financing
- Debtor
- Depreciation
- Equity financing
- Equivalent annual rate
- Equivalent ratios
- Financing
- Future value of an annuity due
- Future value annuity factor
- Inverse proportion
- Investing
- Investment
- Markup interest
- Ordinary annuity
- Percentage
- Present value of ordinary annuity
- Principal
- Proportion
- Rate
- Ratio
- Saving
- Savings accounts
- Simple interest
- Straight-line depreciation
- Term of loan
- Tax

# Introduction

Mathematics applications in business are designed to provide a sound introduction to the uses of mathematics in business and personal finance applications. It has dual objectives of teaching both mathematics and financial literacy. The unit wraps each skill or technique it teaches in a real-world context that shows you the reason for the mathematics you're learning.

This unit has four sections. The first section deals with the concept of ratio, rate, proportion, and percentage. Here, you will see how these concepts are implemented in business. The second section deals with the time value of money and computation of simple and compound interests, annuity, amortization and depreciation of a fixed asset. The third section deals with the concepts of saving, investing, and borrowing money. The last section addresses taxation and the different types of taxes commonly implemented in Ethiopia. Each section accommodates solving problems that are associated with business activities and uses spreadsheets for selected computations.

# 5.1 Basic Mathematical Concepts in Business

The concepts of ratio, rate, proportion and percentage are widely used whenever we deal with business in our daily life activities.

In this section, you will learn how ratios, rates, proportions and percent, are used in many everyday problems.

**Activity 5.1**

When S Super market offers a 25% discount on vegetables and Y Super market advertises "Buy four, get one free". The price-conscious shopper must decide which one is the better offer if she intends to buy five packages of vegetables.

## 5.1.1 Ratio

A ratio, which is a comparison of two numbers by division, is the quotient obtained if the first number is divided by the second number. Since a ratio is the quotient of two numbers divided in a definite order care must be taken to write each ratio in intended order.

For example, the ratio of 6 to 1 is written

$$\frac{6}{1} \text{ (as a fraction) or } 6:1 \text{ (using a colon ).}$$

In general, the ratio of $a$ to $b$ can be expressed as

$$\frac{a}{b} \text{ or } a \div b \text{ or } a:b$$

To find the ratio of two quantities, both quantities must be expressed in terms of the same unit of measures before their quotient is determined.

For example to compare the value US dollars to Japanese Yen, we first convert Yen to dollars and then find the ratios, which is 1:103. Therefore, 1 dollar is worth 103 times as much as Japanese Yen. The ratio has no unit of measurement.

## Example 1

What is the ratio of 2 miles to 1 kilometer?

### Solution

To compare two measurements in different units you must change one of the units of measurement to the other unit or both to another standard unit.

1 mile is equal to 1.6 km. So, 2 miles =3.2 km

Therefore, the ratio is 3.2:1 or 1mile =1600meter and 1km=1000meter;

The ratio of 2 miles to 1 kilometer becomes $\frac{3200}{1000} = \frac{16}{5}$ *or* $16:5$ *or* $3.2:1$.

A ratio can be expressed in one of two ways:

i. Part-to-whole ratio or ii. part-to-part ratio

## Example 2

The following table gives the number of teachers in a given school according to their education level and sex.

a. What is the ratio of female diploma holders to the number of teachers in the school?

b. What is the ratio of diploma holders to degree holders in the school?

| | Diploma Holders | Degree Holders | Total |
|---|---|---|---|
| Male | 26 | 46 | 72 |
| Female | 16 | 12 | 28 |
| Total | 42 | 58 | 100 |

## Solution

a. The first question is asking the part-to-whole ratio, hence it is 16:100 or 4:25.

b. The second question is asking the part-to-part ratio, hence it is 42:58 or 21:29.

### Activity 5.2

In Ethiopia, bank branches to population ratio was 1:83195 in 2010. But this ratio has been changed to 1: 17732 in 2019. Discuss if this ratio has changed.

### Exercise 5.1

1. There are a total of 200 pupils in a certain art school. From those 25 students are major in music and the remaining major in sport. What are the ratio of the students major in music and sport?
2. A profit of Birr 19,560 is to be divided between two partners in the ratio of 3: 1. How much should each receive?
3. Two individuals' shares in a certain company are in the ratio is 10: 1. How many times is the number of shares owned by the major shareholder greater than the number owned by the minor shareholder?

## Equivalent ratios

When a ratio of two numbers such as $\frac{24}{16}$ is a fraction, we divide the numerator and denominator by the same nonzero number to find equivalent ratio. For example,

$\frac{24\div2}{16\div2}=\frac{12}{8}$ $\frac{24\div4}{16\div4}=\frac{6}{4}$ $\frac{24\div8}{16\div8}=\frac{3}{2}$

If a ratio is expressed in simplest form, then both terms of the ratio are whole numbers. But when there is no whole number other than 1 that is a factor of both of these terms. Hence, $\frac{24}{16}$ in simplest form is $\frac{3}{2}$.

**Continued Ratio**

**Activity 5.3**

In concrete the ratio of cement, sand and gravel is 1:3:2. Assume cement is measured in a bag of 50kg, sand and gravel measured in cubic meter. What does this ratio mean? Discuss in a group.

Comparisons can also be made for three or more quantities. A continued ratio is a comparison of three or more quantities in a definite order.

The length of rectangular box is 75 centimeters, the width is 60 centimeters and the height is 45 centimeters. The ratio of length to width is $75:60$ and width to height is $60:45$. Then we can write these ratios as continued ratio $75:60:45$. Here, the measure of length(l), width(b) and height(h) of the rectangular box in simplest form is $5:4:3$.

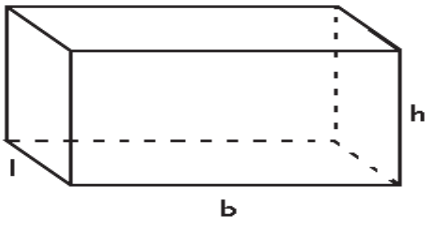



Figure 5.1

**Remark:** In determining the ratio of certain parameters unit of measurement must be the same.

**Example 3**

Allocate Birr 1500 in the continued ratio $2:3:7$.

## Solution

Note that the terms in the ratio are positive integers. First, you need to determine the total number of parts to be allocated. That is $2 + 3 + 7 = 12$.

Now determine the value of each single part, which is obtained by dividing the total amount by the total parts to be allocated: $\frac{1500}{12} = 125$ Birr.

To allocate, multiply each term of the ratio by the value of the single part, i.e.

$2 \times 125 = 250,\ 3 \times 125 = 375,$ and $7 \times 125 = 875.$

Therefore, money will be shared in the quantities Birr 250, Birr 375, and Birr 875, respectively.

## Example 4

The ratio of the angles of a quadrilateral are in the ratio, $\mathbf{7:9:10:10.}$ What is the measure of the angles of a quadrilateral?

### Solution

Since the sum of the angles of a quadrilateral is $360^0$. By adding $7 + 9 + 10 + 10 = 36.$ Dividing 360 by 36 we get 10. It follows that the angles of the quadrilateral are $70^0, 90^0, 100^0, 100^0.$

## Example 5

The Grand Ethiopian Renaissance Dam (GERD) which has a capacity of 74 billion metric cube contains 18.4 billion cubic meters of water in two round filling. That is, 4.9 billion cubic meters in the first round filling and 13.5 billion cubic meters in the second round filling as of July 2021.

a. Find the ratio of the water filled in the dam in the first round to the total capacity of the dam.
b. Find the ratio of the water filled in the dam in two rounds to the total capacity of the dam.
c. How full the dam is?

Figure 5.2. GERD in Guba, Ethiopia, on February 19, 2022.

## Solution

Let $F_1$ be the first round filling, $F_2$ is second round filling, and $F$ is the cumulative in the first $n$ round filling. Setting C is the total capacity

a. $F_1 = 4.9$ Billion cubic meter, $C = 74$ billion cubic meters, $F_1 : C$ implies that 4.9:74. In the first round 1/15 the dam was filled

b. $F = F_1 + F_2 = 18.4$ Billion cubic meters, $F : C$ implies $18.4 : 74$. In the first two rounds about one fourth of the total capacity filled by water.

c. The ratio of 18.4:74 is approximately 1:4, therefore, one quarter of the dam is full

## Exercise 5.2

1. A sum of money was divided between Debora, Kalid, and Mesfin in the ratio of $5 : 3 : 1$, respectively. Debora has received Birr 3504. How much money was there to start with?
2. A block of wood whose length is twice as its width and its height is one-half its width. Write, in the lowest term, the ratio of length to width to height.
3. Express each of the following ratio in the simplest form:
   a. $1\frac{1}{2}$ hour to 30 minutes.
   b. 6 Birr to 50 cents
   c. 1 kilometer to 1 meter

4. A legendary athlete Haile Gebresilassie ran twelve 10,000m international games and won 7 gold, one silver, 2 bronze and certificates for one $5^{th}$ and one $6^{th}$ places between 1992 to 2008. What is the ratio of the number of games he won Gold to the number of games he ran?
5. A chemist wishes to make 15 liters of an acid solution by using water and acid in the ratio 3:2. How many liters of water and acid does she use?

## 5.1.2 Rates

**Activity 5.4**

A combine harvester machine can harvest three hectares of wheat field in one hour at a rate of 150 Birr per hour. If a farmer owns 16.5 hectares of wheat field how much does he pay to harvest his wheat?

A rate, like a ratio, is a comparison of two quantities. But if the quantities have different units of measures, their ratio has a unit of measure.

For Example, if an automobile travels 60 kilometers in 1 hour, its rate of speed is a ratio that compares the total distance covered to the time that the automobile took to travel.

$$\text{rate} = \frac{\text{distance traveled}}{\text{time taken}} = \frac{60}{1}\text{km/h} = 60\ \text{km/h}$$

A rate that has a denominator of 1 is called a unit rate. A rate that identifies the cost of an item per unit is called the unit price. For Example, 26 Birr/kg of sugar or 50 Birr/liter of Milk are unit prices.

**Example 1**

Adane scored 12 goals in 4 Ethiopian premier league games. Express, in lowest terms, the average rate of the number of goals Adane scored per games.

## Solution

$$\text{rate} = \frac{12 \text{ goals}}{4 \text{ games}} = \frac{3 \text{ goals}}{1 \text{ game}} = 3 \text{ goals per game}$$

Adane scored an average rate of 3 goals per game.

In dealing with business, production, population size, and so on, it is common to describe the amount in which certain quantity has increased or decreased based on some starting point or fixed level. This will the rate of change of a given quantity given by:

$$\text{rate of Change} = \frac{\text{amount of change}}{\text{original amount}} = \frac{\text{final amount} - \text{original amount}}{\text{original amount}}$$

The rate of change be a rate of **increase** if the amount of change is **positive** and a rate of **decrease** if the amount of change is **negative**.

### Example 2

The average price of Yirgacheffe coffee for November, 2018/19 was about 1,277 Birr per 17kg (Feresula). In the 2019/20 harvest season (November), Yirgacheffe coffee was traded on average of about 1,780 Birr per 17kg (Feresula). What is the rate of change in the price of one kilogram of coffee from 2018/19 to 2019/20 harvest season?

## Solution

We are given that the original price = Birr 1277 per 17 kg and the new price = Birr1780 per 17 kg. Hence change in price = Birr 1780 – Birr 1277 = Birr 503 per 17 kg

$$\text{rate of change} = \frac{\text{amount of change}}{\text{original amount}} = \frac{503}{1277} = 0.394.$$

The rate change of the price of coffee is 0.394 on the given time interval.

## Example 3

Suppose the daily income a bakery gets from the sale of a special bread increases from Birr 120,000 to Birr 171,000. At the same time the number of special bread sold increases from 1000 to 2000. What is the average change in income per additional special bread that was sold?

### Solution

:Average Rate of Change $= \frac{\text{amount of change}}{\text{change in unit}} = \frac{171000-120000}{2000-1000} = \frac{51000}{1000} = 51$

So the average increase in revenue, per additional bread sold, is Birr 51.

## Exercise 5.3

1. Yohannes and Semira use computers for word processing. Yohannes can type 1215 words in 15 minutes while Semira can type 2700 words in 30 minutes. Who is the faster typist?
2. A carpenter's daily production of chairs increased from 20 units to 40 units. At the same time his daily income (or revenue) increased from 1600 Birr to 2400 Birr. As a result of this increase, what is the rate of change of income per unit?
3. A steel company imported 70 tons of raw materials from India in 2010. In 2014 the company imported 105 tons of raw materials from the same country. What is the rate of change of amount imported?

## 5.1.3 Proportion

A **proportion** is an equation that states that two ratios are equal. Since the ratio $5:30\ or\ \frac{5}{30}$ is equal to the ratio$1:6\ or\ \frac{1}{6}$, we may write the proportion

$$5:30 = 1:6 \text{ or} \frac{5}{30} = \frac{1}{6}.$$

Each of these proportion reads as " 5 is to 30 as 1 is to 6." For $a, b, c, d \in \mathbb{R}$, with $b \neq 0$ and $d \neq 0$, one way of denoting a proportion is $a : b = c : d$, which is read as "$a$ is to $b$" is equal to "$c$ is to $d$."

Of course, by definition, $\frac{a}{b} = \frac{c}{d}$ means that a proportion is an equation between two ratios. In the proportion, $a : b = c : d$, with $b \neq 0$ and $d \neq 0$, the four numbers are referred as the **terms** of the proportions. The first and the last terms $a$ and $d$ are called **the extremes**; the second and third terms $b$ and $c$ are called the **means**. In the proportion, $a:b = c:d$, the product of the extremes is equal to the product of the means; that is,

$\frac{a}{b} = \frac{c}{d}$ is equivalently represented as $a \times d = b \times c$.

For three quantities $a, b$ and $c$ such that $\frac{a}{b} = \frac{b}{c}$ which is equivalent to $b^2 = ac$ is called the **mean proportional** between $a$ and $c$.

## Example 1

On a residence house construction plan of Ato Admasu, 1 cm on the paper represent 25 meters on the ground. Find the distance on the ground such that the distance represented by 3.20 cm on the paper.

### Solution

On the map we have the ratio 1cm to 25m. Converting meter to centimeters, 25m=2500cm. Let $x$ be the distance on the ground. Then the distance represented by 3.20 cm on the plan can be found by the proportion

$$\frac{1\text{cm}}{2500\text{cm}} = \frac{3.2\text{cm}}{x},$$

$$x = \frac{3.2\text{cm} \times 2500\text{cm}}{1\text{cm}} = 8000 \text{ cm}.$$

Hence, 3.2 cm on the paper is equivalent to 80m on the ground.

**Compound proportion**

From the above discussion you have seen how change in the quantity of one variable depends on a change in the quantity of another variable (i.e., simple proportion). However, the value of a quantity of the variable most often depends on the value of two or more other variable quantities. For example,

- The cost of sheet metal depends on the area of the sheet, thickness of the sheet, and the cost per unit area of the metal.
- The amount of interest obtained depends on the amount of money deposited in a bank, length of time it is deposited, and rate of interest per year.
- The amount of product produced depends on the amount of capital and labor hour units used.

**Definition 5.1**

A compound proportion is a situation in which the quantity of one variable depends on two or more other quantities of other variables. Specifically, if a variable quantity $y$ is proportional to the product of two or more variable quantities, we say that $y$ is jointly proportional to these variable quantities, or $y$ varies jointly as these variables.

Two quantities $x$ and $y$ are said to be in direct proportion if they increase or decrease together in such a manner the ratio of their corresponding values remains constant. That is, $\frac{x}{y} = k$, for some positive constant $k$. Hence, $k$ is a constant of variation.

Two quantities $x$ and $y$ are said to be an inverse proportion if an increase in $x$ causes a proportional decrease in $y$ and vice-versa, in such a manner that the product of their corresponding values remains constant. That is, $xy = k$ where $k$ is a positive number, if $x$ and $y$ are in inverse proportion.

When two quantities $x$ and $y$ are in direct proportion (or vary directly), they are written as $x \propto y$. Symbol " $\propto$ " stands for 'is proportional to'. When two quantities $x$ and $y$ are in inverse proportion (or vary inversely) they are written as $x \propto \frac{1}{y}$.

## Example 2

Under the condition that the temperature remains constant, the volume of gas is inversely proportional to its pressure. If the volume of gas is $4m^3$ at a pressure of 1 kilopascal, then what will be the pressure of the gas, if its volume is $3m^3$ at the same temperature?

### Solution

By Boyle's law, pressure and volume of a gas are inversely proportional at constant temperature. Let the required pressure be $x$.

| Volume of gas in meter cube | 4 | 3 |
|---|---|---|
| Pressure of gas in kilopascal | 1 | $x$ |

The relationship for Boyle's Law can be expressed as follows: $P_1V_1 = P_2V_2$, where $P_1$ and $V_1$ are the initial pressure and volume values, and $P_2$ and $V_2$ are the values of the pressure and volume of the gas after change.

So $4 \times 1 = 3 \times x$, Therefore, $x = 4/3 = 1.33$. The volume decreased to 3cubic meter then the pressure increased to 1.33 kilopascal.

## Example 3

The table gives pairs of values for the variables $x$ and $y$.

| $x$ | 1 | 2 | 3 | 7 | ? |
|---|---|---|---|---|---|
| $y$ | 9 | 18 | 27 | ? | 900 |

a. Show that one variable varies directly with the other variable.

b. Find the constant of variations by comparing $y$ to $x$.

c. Express the relationship between the variables in a formula.

d. Find the values missing in the table.

## Solution

$\frac{x}{y} = \frac{1}{9}$ $\frac{x}{y} = \frac{2}{18} = \frac{2\div 2}{18\div 2} = \frac{1}{9}$ $\frac{x}{y} = \frac{3}{27} = \frac{3\div 3}{27\div 3} = \frac{1}{9}$ .

a. Since all the given pairs of values have the same ratio, x and y vary directly.

b. Constant of variation $= \frac{9}{1} = 9$.

c. Write by including the constant of variation 9. That is, $\frac{y}{x} = 9$, $y = 9x$.

d. Solve the equation in c. when $x = 7, y = 9 \times 7 = 63$, when $y = 900$,

$$900 = 9x,$$

$$x = \frac{900}{9} = 100.$$

## Exercise 5.4

1. Both $x$ and $y$ vary directly with each other. Let $x$ is 10, $y$ is 15, then, write four corresponding pairs of values of $x$ and $y$.
2. Both $x$ and $y$ vary inversely with each other. When $x$ is 10, $y$ is 6. Write four corresponding pairs of values of $x$ and $y$.
3. If the price of 15 postal stamps is Birr 60, then what is the price of 72 postal stamps?
4. If a deposit of Birr 2,000 earns an interest of Birr 500 in 3 years, how much interest would a deposit of Birr 36,000 earn in 3 years with the same rate of simple interest?
5. The mass of an Aluminum rod varies directly with its length. If a 16 cm long rod has a mass of 192 g, find the length of the rod whose mass is 105 g.

## 5.1.4 Percentage

### Activity 5.5

Complete the following table without **using** calculators.

| percentage | 250 | 100 | 625 | 2000 |
|---|---|---|---|---|
| 10% | | | | |
| 25% | | | | |
| 53% | | | | |
| 15% | | | | |
| 7.2% | | | | |

Problems dealing with discounts, commissions and taxes involve percentages. A percentage is the numerator of a fraction whose denominator is 100, which is a ratio of a number to 100. The term percent is denoted by % which means "per one hundred". For example, 7% (read as 7 percent) is a ratio of 7 to 100. A percent can be expressed as a fraction or decimal:

$$7\% = \frac{7}{100} = 0.07$$

If an item is taxed at a rate of 15%, then a 100 Birr book costs an additional 15 Birr for tax. Here, three quantities are involved:

- The base, the sum of money being taxed, is Birr 100.
- The rate, or the rate of tax is 15% or 0.15.
- Percentage, or the amount of money being taxed, is 15 Birr.

These three related terms may be written as a proportion: $\frac{percentage}{base} = rate$ or as a formula: $Percentage = base \times rate$

### Example 1

Find the amount of turnover tax on a 400,000 Birr sale when the turnover tax rate is 2%.

## Solution

Let percentage or amount of tax= $base \times rate = 400{,}000 \times 0.02 = 8000$. The amount of tax is 8,000 Birr.

## Example 2

Ethio Telecom offers a discount of 25% off by using Tele Birr purchase of its products. What is a regular price of a product that a customer purchased for 81 Birr?

## Solution

The rate of the discount is 25%. Therefore the customer paid (100-25)% or 75% of the regular price. The percentage is given as Birr 81, the base is unknown. Let n=regular price or base.

Method I: Use proportion:

$$\frac{\text{percentage}}{\text{base}} = \text{rate}$$

$$\frac{81}{n} = \frac{75}{100}$$

$$81 = 0.75n, \quad n = \frac{81}{0.75} = 108$$

Method II: Formula Percentage=base×rate= $n \times 0.75 = 81$,

$n = \frac{81}{0.75} = 108$. Hence, the regular price of the product was 108 Birr.

**Check:** If 25% of 108 is subtracted from 108 does the difference equal 81?

$$0.25(108) = 27$$

$$108 - 27 = 81. \quad \text{(Correct)}$$

## Example 3

What is the meaning of the percentage symbol in road? What angle the percentages correspond to in road sign? See Figure 5.3 and Figure 5.4.

Figure 5.3 Percentage sign in road

Trigonometric in real life.

What angle the percentages correspond to in road sign?

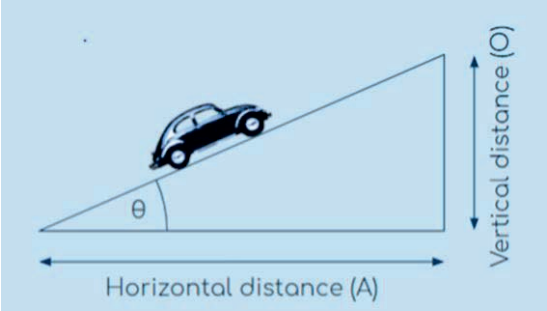



Figure 5.4

## Solution

The sign 10% means that the vertical change is 10% of the horizontal change: that is, for every 10 meter travel horizontally there is 1 meter up. In general a bigger percentage translates as a steeper hill.

Calculating the slope of a straight line, as would be taught in a lower class ("rise over run" or "change in $y$ divided by change in $x$") is equivalent to calculating, $\tan\theta = \frac{O}{A}$. Therefore, 10% means $\tan\theta = 0.1$. $\theta$ is the tangent inverse of 0.1. In our example 10% is the same as an angle $5.71^0C$ as shown in Figure 5.3.

## Exercise 5.5

1. In (i) - (iv), find each indicated percentages.
   i. 2.5% of 400
   ii. 12.5% of 175
   iii. 35% of 1500
   iv. 100% of 77
2. Find each number or base.
   a. 8% of what number is 12?
   b. 75% of what number is 24?
   c. 33% of what number is 200?
   d. 15% of what number is 1.91?
3. How much silver is found in 80kg of an alloy that is 8% silver?

4. Last week, Bontu answered 30 out of 36 questions correctly on a test. This week she answered 20 out of 24 questions correctly. On which test did she have better result?
5. The monthly salary of Mr. Tilahun is 26% more than Mr. Jemal salary. Assume Mr. Jemal salary is Birr 15396.00. What is the salary of Mr. Tilahun?
6. What does 25% mean in road symbol? What is the slope of the road in degree?

## Markup

In order to make a profit, any institution or company must sell its products for more than the product costs the company expended to produce or buy the product. The difference between a product selling price and its cost is called **markup**.

Markup $=$ selling price $-$ cost price.

### Example 4

The price of cement is Birr 250 per quintal. If you sell it for Birr 330 per quintal, what is the markup per quintal?

### Solution

Markup $=$ Selling price $-$ Cost, thus,

Birr 330 per quintal $-$ Birr 250 per quintal $=$ Birr 80 per quintal.

Markup is usually expressed in terms of percentage with respect to selling price and cost.

Markup with respect to selling price is given by;

$$\text{Markup percent} = \frac{\text{markup}}{\text{selling price}} \times 100\%.$$

Similarly markup with respect to cost is given by:

$$\text{Markup percent} = \frac{\text{markup}}{\text{cost}} \times 100\%.$$

## Example 5

A boutique bought a T-shirt for Birr 546 and wanted a markup of 30% on retail. What is the selling price?

### Solution

Given cost = Birr 546. Markup percent is 30% on selling price. Then we need to find selling price.

Cost = selling price – markup $= (100 - 30)\% = 70\ \%$ of selling price.

This is called the complement of markup percent on selling price.

Hence, the selling price will be cost = $0.70 \times selling\ price$

$$\Rightarrow 546\ Birr = 0.70 \times selling\ price$$

$$\Rightarrow selling\ price = \frac{546}{0.7} = 780\ Birr.$$

## Discount

### Activity 5.6

Sometimes retailers and manufacturing company offer a discount. Why do those companies offer a discount?

**Discount** is a reduction on the original selling price. **Regular price** is the price at which an article is offered for sale. **Sale price** is the price that the customer pays in the case of a discounted article. The sale price is obtained by deducting the discount from regular price.

## Example 6

In Ethiopian New Year holiday Ethio telecom offer 20% discount on international voice call. The regular (original) price is Birr 6.00 per minute. Calculate the call price.

## Solution

call price = regular price − discount $= 6 - 0.2 \times 6 = 6 - 1.2 =$ *Birr* 4.80 per minute.

$$Discount\ rate = \frac{disount}{regular\ price} \times 100\%$$

### Example 7

On festive shopping the price of edible oil is 20% discount of Birr 175.00 per liter. Calculate the original price.

## Solution

To calculate original price by using discount rate formula,

$$Discount\ rate = \frac{discount}{regular\ price} \times 100\%$$

$$0.2 = \frac{discount}{regular\ price} = \frac{175}{regular\ price}$$

$$regular\ price = \frac{175}{0.2} = 875$$

Hence, the original price of the edible oil is Birr 875.00 and its selling price is Birr 700.00.

### Exercise 5.6

1. A car dealer reduces the price of last year's models at the clearance sale of the end of the year, by a certain amount. If a certain four-door model has been sold at a discounted price of Birr 510,000, with a discount of Birr 90,000, what is the percentage of the discount?
2. One year, a furniture manufacturing firm made a profit of 520,000 Birr. This represented 8% of the volume of the business for the year. What was the volume of the business in that year?

3. A proposal made in the regional council members to raise the minimum wage from Birr 12 to Birr 15 an hour. What was the proposed percent of increase to the nearest tenth of a percent?
4. A pair of shoes that costs Birr 110 is sold for Birr 155. Find the markup and the percent of the markup based on the retail (selling price).
5. What is the percent markup on cost if the markup on retail is 37%?
6. Abebe sold a quintal of Teff at Birr 5250 with 45% markup on selling price. Find the cost.
7. Find the percent markup on cost, if markup of the selling price is 30%.
8. Calculate the discount and selling price for a Birr 1700.00 wheat offered at 18% discount.
9. Calculate the regular price of a television set, assuming that a 25% discount would have been worth Birr 2,000.00.

## 5.2 Time Value of Money

For living in someone's house or using properties, individuals return the cost of the service they were given in terms of money. The money that they pay in return is known as **rent.**

Now, suppose someone needs to borrow Birr 200, and you agree to lend him. If he offered to pay you back the full Birr 200 for one year, would you agree such a term? It hardly seems fair because you wouldn't get any other compensation. Just as in renting the house, even though you will eventually get your property back, it only seems fair if you get some benefit for allowing the use of what belongs to you.

We ordinarily call the payment for the temporary use of property such as houses, apartments, equipment, or vehicles a rent though we don't normally use that term in the case of money. Instead we call that payment for use of money an interest. Interest is either simple or compound.

## 5.2.1 Simple Interest

**Definition 5.2**

Interest is what a borrower pays a lender for the temporary use of the lender's money. In other words, it is the "rent" that a borrower pays a lender to use the lender's money.

Interest is paid in addition to the repayment of the amount borrowed. In some cases, the amount of interest is spelled out explicitly. If we need to determine the total amount to be repaid, we can simply add the interest on to the amount borrowed.

From the investment point of view, interest is income from invested capital for investor and expense for the one who used the money or for the borrower. The capital originally invested is called the principal. The principal of a loan is the amount borrowed. The sum of the principal and interest due (or paid) is called the amount (or future value or accumulated value).

There are a few special terms that are used with loans as well.

A **debtor** is someone who owes someone else's money. A **creditor** is someone to whom money is owed. The amount of time for which a loan is made is called its **term**.

For simple interest, the interest is computed on the principal during the whole time, or term of the loan, at the stated annual rate of interest.

To calculate simple interest, we multiply the amount borrowed times the interest rate (as a decimal) times the amount of time.

$$I = Prt \qquad 5.1$$

where, $I$ simple interest, $P$ the principal, $r$ annual interest rate and $t$ term of loan.

**Example 1**

Ashamo borrows Birr 18,500 at 9.5 % simple interest rate for 2 years. How much interest will he pay?

## Solution

The principal $P = 18{,}500$, the interest rate $r = 9.5\% = \frac{9.5}{100} = 0.095$ and the time t= 2 years. So, we begin with our formula $I = Prt = 18500 \times 0.095 \times 2 = 3515$. The amount of interest paid in two years of time is Birr 3515. The total amount of money after two years will be

$$A = P + I = 18500 + 3515 = 22015$$

This amount is known as future value of a simple interest.

**The future value of a simple interest**

Beginning: $A = P$

At the end of first year: $A = P + I = P + Pr = P(1 + r)$

At the end of second year: $A = P(1 + r) + Pr = P(1 + r + r) = P(1 + 2r)$

Similarly at the end of third year; $A = P(1 + 3r)$.

Finally, after $t$ years the future value of a simple interest $A$ is

$$\boldsymbol{A = P(1 + rt)},$$

where $P$ is the principal, $r$ is the simple interest rate per year, and $t$ is the time in years.

Sometimes interest is paid in situations we might not normally think of as loans. If you deposit money in a bank account, you probably expect to be paid interest even though you probably don't think of your deposit as a loan. In reality, though, it actually is a loan. When you are depositing money to a bank account you are actually loaning that money to the bank.

### Example 2

If Birr 2,500 is deposited in Commercial Bank with a simple annual interest rate of 7 %, find the amount of the interest and future value at the end of the fourth month.

## Solution

In this example $P = 2500, r = 0.07, t = \frac{4}{12} = 0.33\ year.$ Substituting these values in equation (5.1) we obtain,

$$I = Prt = 2500 \times 0.07 \times 0.33 = 57.75$$

The amount of interest will be Birr 57.75. The future value A is

$$A = P + I = 2500 + 57.75 = 2557.75.$$

There are different types of bank deposits. Checking and savings accounts are familiar examples of ways in which you loan money to banks. These are sometimes referred to demand accounts because you can withdraw your money at any time you want (i.e., "on demand"). Another common type of account is a certificate of deposit, or CD or commonly called "fixed time deposit." When you deposit money into a CD, you agree to keep it on deposit at the bank for a fixed period of time.

### Example 3

A Saving and Credit Association deposited Birr 13,000,000.00 (Thirteen million) into a 1-year fixed time deposit paying with 13.5% simple interest per annum. How much will his account be worth at the end of the term?

## Solution

$$I = Prt = 13000000 \times 0.135 \times 1 = 1{,}755{,}000$$

The Saving and Credit Association gets Birr 1,755,000 interest yearly. At the end of the term, the Saving and Credit Association account will contain both the principal and interest, so the total value of the account will be

$13{,}000{,}000 + 1{,}755{,}000 = 14{,}755{,}000$ .

## Exercise 5.7

1. Derartu loaned Tesfaye Birr 12,000 for 6 months. Tesfaye paid back Birr 12,600. How much interest did he didrpay? What is the annual interest rate?
2. Gemechu loaned Hagos Birr 800, and 2 years later Hagos will pay back Birr 900. How much total interest will Gemechu receive? What is the annual interest rate?
3. Samir borrowed Birr 7,829.14 for 1 year at a simple interest rate 9.75% annually. How much will he need to repay the loan?

## Determining Principal, Interest rate and Time

So far, we have developed the ability to calculate the amount of interest due when we know the principal, rate and time. However, situations may arise where we already know the amount of interest, and instead need to calculate one of the other quantities. For example, consider these situations:

- A retiree hopes to be able to generate Birr 10,000 income per month from an investment account that earns 12 % simple interest. How much money would he need in the account to achieve this goal?
- Tekletsadik borrowed Birr 1500 from his brother-in-law, and agreed to pay back Birr 1650 one year later. What rate of simple interest is Tekletsadik paying for this loan?
- Zehara deposited Birr 9,750 in a savings account that pays 5.25% simple interest per year. How long will it take for her account to grow to Birr 10,000.00?

The first bulleted question is **finding principals**. For the problem monthly income would be $Income = 10{,}000$, rate of interest $r = 12\%$, the time is one month $=\frac{1}{12}$ of a year.

From the Formula, $I = Prt$, dividing both sides by $rt$, we obtain

$P = \frac{I}{rt} = \frac{10000}{0.12\times(\frac{1}{12})} = 1{,}000{,}000.00$ to earn Birr 10,000 monthly. The retiree must invest Birr 1 million on the stated interest rate.

On the second bulleted problem, principal, future value of money and the term of the loan were given. The required is the **rate of simple interest**.

Similar to the above problem rate of interest is calculated by the formula $r = \frac{I}{Pt}$,

I=1650-1500=150, which is the difference between pay back amount and borrowed amount.

$$r = \frac{150}{1500 \times 1} = 0.1.$$

So the rate is 0.1 or 10%.

On the third bulleted problem is finding the term of the loan or time needed to settle the loan.

$$P = 9750, I = 10000 - 9750 = 250, r = 5\frac{1}{4}\% = 0.0525$$

By dividing the formula $I = Prt$, both sides by $Pr$, we obtain

$$t = \frac{I}{Pr}$$

Therefore, $t = \frac{250}{9750\times0.0525} = 0.4884$, the number tell us 0.4884 years, which is less than half a year. Multiply by 365 days and we obtain

$0.4884 \times 365 = 178.26$ days. It takes around 179 days to grow to Birr 10,000.

## Example 4

Dakito borrowed Birr 2,000 and paid back a total of Birr 2,125 in a year. How much interest did he pay? What is the annual interest rate?

### Solution

Dakito paid an interest I, which is the difference between principal and returned amount, i.e. $I = 2{,}125 - 2{,}000 = 125.$

From Equation, $r = \frac{I}{Pt}$, in this example $P = 2{,}000, I = 125$ and $t = 1$. Therefore, $r = \frac{125}{2000} = 0.0625$. Thus, the annual interest rate is 6.25%.

## Example 5

Find the term of a loan if the principal is Birr 20,000, the interest rate is 8%, and the total interest is Birr 320.

### Solution

Given $P = 20{,}000$, $r = 8\%$, and interest earned is $I = 320$. Using the formula $I = Prt$, we get $t = \frac{I}{Pr} = \frac{320}{20000 \times 0.08} = 0.2$. The term of the loan can be 0.2 year which is equivalent to $0.2 \times 360 = 72$ days or 2.4 months.

## Exercise 5.8

1. Find the total amount that will be required to pay off a 3-year loan of Birr 14,043.43 at 6.09% simple interest per year.
2. Find the principal for a loan if the term is 2 years, the simple interest rate is 9% annually and the interest totals Birr 63.00.
3. How much does Kedija need in her investment account if she wants to be able to receive Birr 450 per month in income from it, assuming a simple interest rate of 7.5% annually?

## 5.2.2 Compound Interest

*"The most powerful force in the universe is compound interest."* **Albert Einstein**

Simple interest is very rarely used in real life: almost all banks and other financial institutions use compound interest. This is when interest is added (or compounded) to the principal sum so that interest is paid on the whole amount. Under this method, if the interest for the first year is left in the account, the interest for the second year is calculated on the whole amount so far accumulated.

To begin with, let's consider a Birr 6,000 loan for 5 years at 8% simple interest.

From your previous work you can easily calculate the interest:

$I = Prt = 6000 \times 0.08 \times 5 = 2400.$

And so the maturity value would be $6000 + 2400 = 8400\ Birr.$

This calculation takes us from principal to maturity value without any thought about how the amount of interest grows over the loan's term.

Suppose, instead of just jumping from the Birr 6,000 principal to the Birr 8,400 maturity value 5 years later, let us take a look at the loan year by year along the way.

In the first year, the interest would be

$$I = Prt = 6000 \times 0.08 \times 1 = 480.$$

The principal is still Birr 6,000, the interest rate is still 8%, and the second year is 1 year long, just like the first one. Likewise, the interest in the second, third, fourth, and fifth years would also be the same. Thus under simple interest the loan is growing at a constant rate of Birr 480 per year.

For the case of compound interest, at the end of the first year

$I = \Pr t = 6000 \times 0.08 \times 1 = 480.00$.

The total amount, $A = P + I = 6000 + 480 = 6480$.

At the end of second year, $I_2 = 6480 \times 0.08 \times 1 = 518.40$. As shown in Table 1 below the interest is different yearly. For interest at the end of each years summarized below in Table 1.

Table 1 Simple and Compound interest of Birr 6000 at rate 8% annually

| Simple interest of Birr 6000 at rate 8% annually | | | Compound interest on Birr 6000 at rate 8% annually | | |
|---|---|---|---|---|---|
| **Year** | **Interest** | **Balance** | **Year** | **Interest** | **Balance** |
| start | Not applicable | 6000 | start | Not applicable | 6000 |
| 1 | 480 | 6480 | 1 | 480 | 6480 |
| 2 | 480 | 6960 | 2 | 518.4 | 6998.4 |
| 3 | 480 | 7440 | 3 | 559.87 | 7558.27 |
| 4 | 480 | 7920 | 4 | 604.66 | 8162.93 |
| 5 | 480 | 8400 | 5 | 653.03 | 8815.97 |

## Activity 5.7

a. Show how to calculate simple interest and balance in spreadsheet.

b. Show how to calculate compound interest and balance in spreadsheet.

As the balance grows year by year, so does the interest. It is worthwhile to compare the growth of the account year by year under compound interest versus its growth under simple interest.

Table 2 Comparing compound and simple interest of Birr 6000 deposited at a rate of 8% annually.

| | Simple Interest | | Compound Interest | | | |
|---|---|---|---|---|---|---|
| Year | Interest | End Balance | Year | Interest | End Balance | Difference in end balance |
| start | Not applicable | 6000 | start | Not applicable | 6000 | 0 |
| 1 | 480 | 6480 | 1 | 480.00 | 6480.00 | 0.00 |
| 2 | 480 | 6960 | 2 | 518.40 | 6998.40 | 38.40 |
| 3 | 480 | 7440 | 3 | 559.87 | 7558.27 | 118.27 |
| 4 | 480 | 7920 | 4 | 604.66 | 8162.93 | 242.93 |
| 5 | 480 | 8400 | 5 | 653.03 | 8815.97 | 415.97 |

**Note** that, as time goes by and the account balance grows, so does the impact of compounding. Over the first couple of years, the difference between simple and compound interest is not significant. But as the principal used for the compound interest grows larger and larger, the interest earned on that principal also grows more and more.

## Example 1

You deposit Birr 3000.00 in a high-earning account paying 9% compound interest yearly and leave it for three years. What will be the balance on the account at the end of that time?

## Solution

Balance after 0 years Birr 3000.00

Interest: 9% of 3000.00 Birr=$0.09 \times 3000 = 270.00$

Balance after 1 year: Birr 3,270.00

Interest: 9% of 3270.00=$0.09 \times 3270 = 294.30$,

Balance after 2 years: Birr 3,564.30.

Interest: 9% of 3564.30=$0.09 \times 3564.3 = 320.78$.

Balance after 3 years: Birr 3,885.08. At the end of third year the depositor will get a compound interest of Birr 885.08.

### 5.2.2.1 Compound Interest Formula

It should already be clear that for long periods, the year-on-year method of calculating compound interest is somewhat cumbersome, but fortunately there is a formula.

If an initial deposit of $P$ is made at time zero with annual rate $r \times 100\%$, then after the first period the value of the account is $A_1 = P + rP = P(1 + r)$, after the second period the value is

$A_2 = A_1 + rA_1 = A_1(1 + r) = P(1 + r)^2$, and so forth. In general, the value of the account at time $n$ is

$$A_n = P(1 + r)^n \qquad 5.2$$

**Example 2**

Use the compound interest formula to find how much Birr 5,000 will grow to in 5 years at 8% annual compound interest. What will happen in 50 years?

## Solution

After 5 years Birr 5000 became an amount. Substituting

$$A = P(1 + \frac{8}{100})^5$$

$$A = 5{,}000(1 + 0.08)^5$$
$$A = 5{,}000(1.08)^5 = 5000 \times 1.469 = 7{,}345.$$

The total compound interest of the period computed by subtracting $P$ from $A$. That is,

$$A - P = P(1 + \frac{r}{100})^n - P = P[\left(1 + \frac{r}{100}\right)^n - 1].$$

Therefore, compound interest (C.I) is

$$C.I = P\left[\left(1 + \frac{r}{100}\right)^n - 1\right]. \qquad 5.3$$

The interest to be paid in 5 years on Birr 5,000 is $A - P$=7345-5000=2,345. Birr.

In 50 years simply by substitution you get

$$A_{50} = 5000(1 + \frac{8}{100})^{50} = 234508.063.$$

## Example 3

Calculate the number of complete years in which a sum of money deposited at 20% compound interest will be doubled.

### Solution

Let $P$ be the original amount deposited after $n$ years. This amount grows to doubled, i.e., $2P$.

$A = P(1 + \frac{20}{100})^n = 2P$. Dividing both sides by $P$ you obtain

$(1 + 0.2)^n = 2$. To solve for $n$ you can use your knowledge of logarithmic function.

$$n \log 1.2 = \log 2$$
$$n = \frac{\log 2}{\log 1.2} = 3.8$$

The deposit must be left for 3.8 years but as interest is paid yearly, it would have to be left for 4 years.

## Exercise 5.9

1. You deposited Birr 10,000.00 in a high-earning account paying 9.5% compound interest and leave it for five years. What will be the balance on the account at the end of that time? What is the compound interest of the period?
2. For how long must a sum be deposited in an account paying 14% compound interest in order to triple in value?
3. Calculate the annual rate of compound interest that will allow a principal sum to double in value after five years.
4. Calculate the principal which, if deposited at 9.5% compound interest, will grow to Birr 4000 after three years.
5. Ayantu deposited Birr 2,000 at 8% interest compound annually. How many years will it take her to get Birr 3,000?

### 5.2.2.2 Compounding Frequencies

**Activity 5.8**

In a group collect information from Bank professionals and savers about:

- How many times in a year does Commercial Bank of Ethiopia calculate interest on deposit of money?
- Choose any bank or credit union and ask how many times they pay interest on annual deposit.

Interest is usually compounded more than once a year. The quoted rate of interest per year is called annual or nominal rate and the interval of time between successive interest calculations is called conversion period or compound period.

As you looked, in previous example, there was no difference between simple and compound interest at the end of the first year. Since no interest was credited to those accounts until the end of the first year, there was no opportunity for interest on interest until the end of the first year. In the second year, interest on interest began to

make a difference, and the difference became more pronounced the following year. In this case, interest on interest combined with interest on interest make a difference significantly. What will happen if the interest is calculated monthly or even daily?

What's more, over the course of each year compounding would take place in a total of 12 times, rather than just once. But how much more interest would this monthly compounding mean? If you invested Birr 6,000 at 8% compounded annually, when the first interest is credited at the end of the first year you would have Birr 6,480. Now suppose that we look at the same Birr 6,000 at the same 8%, but this time we compound the interest every month.

The interest earned for the first month would be:

$I = Prt = 6000 \times 0.08 \times \left(\frac{1}{12}\right)$ =Birr 40.

So at the end of the first month the account balance would be Birr 6040.00.

Following the same approach, we find that the next month's interest works out to Birr 40.27—a bit more and continuing on through the rest of the year, we get the result given in Table 3.

Table 3: Compound interest on Birr 6000 compounded monthly at a rate 8%.

| Month | Beginning Balance | Interest | Ending Balance |
|---|---|---|---|
| 1 | 6,000.00 | 40.00 | 6,040.00 |
| 2 | 6,040.00 | 40.27 | 6,080.27 |
| 3 | 6,080.27 | 40.54 | 6,120.80 |
| 4 | 6,120.80 | 40.81 | 6,161.61 |
| 5 | 6,161.61 | 41.08 | 6,202.68 |
| 6 | 6,202.68 | 41.35 | 6,244.04 |
| 7 | 6,244.04 | 41.63 | 6,285.66 |
| 8 | 6,285.66 | 41.90 | 6,327.57 |
| 9 | 6,327.57 | 42.18 | 6,369.75 |
| 10 | 6,369.75 | 42.47 | 6,412.22 |
| 11 | 6,412.22 | 42.75 | 6,454.96 |
| 12 | 6,454.96 | 43.03 | 6,498.00 |

Let's Calculate

$6000 \times 0.08 = 40.00$

Since $r$ is the annual interest rate and the interest is compounded m times per year, the year is divided into m equal conversion periods and the interest rate during each conversion period is $i = r/m$, that is we get interest r/m every $1/m$ years.

Notice that this is Birr 18.00 more than we had at the end of the first year with annual compounding.

Now, if the interest i3s compounded for $t$ years, then there will be $n = mt$ conversion periods in $t$ years. Thus, if you put $n = mt$ and replace $r$ by the expression of interest rate per each conversion period $\boldsymbol{i} = \frac{\boldsymbol{r}}{\boldsymbol{m}}$, we have the future value of compound interest given by:

$$A = P(1 + \frac{im}{100})^{mt} \qquad 5.4$$

Where $A$ is the amount or future value, $P$ is the principal or present value, $\boldsymbol{i} = \frac{\boldsymbol{r}}{\boldsymbol{m}}$, r is annual or nominal rate, $t$ is time in years, and $m$ is the number of conversion periods per year.

In working with problems involving interest, we use the term of payment periods as follows:

- Annually means once a year, i.e. $m = 1$.
- Semi-annually means twice a year, i.e. $m = 2$.
- Quarterly means four times a year, i.e. $m = 4$.
- Monthly means 12 times a year, i.e. $m = 12$ and.
- Daily means 365 times a year, i.e. $m = 365$.

Collectively the value of $m$ implies compounding frequencies, i.e. how many times in a year the interest is calculated on the given deposited amount of money or investment.

## Example 1

If Birr 100 is deposited in the Commercial Bank with 10% interest rate annually, find the amount if it is compounded annually, semi-annually, quarterly, monthly, weekly and daily at the end of one year. Assume withdrawal or deposit not allowed on the given interval of time.

## Solution

The given values are $P = 100$, at a rate of $r = 10\% = 0.1$ for a period of one year, and compound period of $n = mt$.

a. Annually means $m = 1$, so that the amount at the end of the year is

$$A = 100(1 + \frac{0.1}{1})^1 = 100(1 + 0.1)^1 = 100(1.1) = \text{Birr } 110.$$

b. Semi-annually means $m = 2$, so that the amount at the end of the year is

$$A = 100(1 + \frac{0.1}{2})^2 = 100(1 + 0.05)^2 = 100(1.05)^2 = \text{Birr } 110.25.$$

c. Quarterly means $m = 4$, so that the amount at the end of the year is

$$A = 100(1 + \frac{0.1}{4})^4 = 100(1 + 0.025)^4 = 100(1.025)^4 = \text{Birr } 110.38.$$

d. Monthly means $m = 12$, so that the amount at the end of the year is

$$A = 100(1 + \frac{0.1}{12})^{12} = 100(1 + 0.0083)^{12} = 100(1.0083)^{12}$$
$$= \text{Birr } 110.47.$$

e. Weekly means $m = 52$, so that the amount at the end of the year is

$$A = 100(1 + \frac{0.1}{52})^{52} = 100(1 + 0.0019)^{52} = 100(1.0019)^{52}$$
$$= \text{Birr } 110.51.$$

f. Daily means $m = 365$, for simplified exact method, so that the amount at the end of the year is

$A = 100(1 + \frac{0.1}{365})^{365} = 100(1 + 0.00027)^{365} =$

$100(1.00027)^{365} = \text{Birr } 110.52.$

Bankers use $m = 360$, this sometimes known as Bankers rule

$A = 100(1 + \frac{0.1}{360})^{360} = 100(1 + 0.000277)^{360} =$

$100(1.000277)^{360} = \text{Birr } 110.52.$

These two methods have no significant difference in two decimal places. You can observe that when the time, principal and interest rates are kept fixed and the number of times the interest is compounded increases, the amount will also increase.

## Example 2

Ato Deribe made the following transactions on his account at Bank. Deposited Birr 25,000 on 1st January 2018, withdraw Birr 6,000 on 1st July 2020, deposited Birr 18,000 on 1stJanuary 2019. If the account earns 7.5% interest rate per year compounded monthly, find the balance on the account on 1st January 2022.

### Solution

The solution of this example can be calculated by using spreadsheet as follow. As expressed from January 1, 2018 to July 1, 2020 the deposit amount earned interest for 30 month continuously.

| Date | Amount | Deposit | Withdrawal | Interest earned | Ending Balance |
|---|---|---|---|---|---|
| 1-Jan-18 | 25,000.00 | | | 1,940.81 | 26,940.81 |
| 1-Jan-19 | 26,940.81 | 18,000.00 | | | 44,940.81 |
| 31-Dec-19 | 44,940.81 | | | 3,488.87 | 48,429.69 |
| 1-Jul-20 | 48,429.69 | | 6,000.00 | 1,844.73 | 44,274.41 |
| 1-Jan-22 | 44,274.41 | | | 5,254.51 | 49,528.93 |

Ato Deribe will have Birr 49,528.93 on his account on January 1, 2022.

## Exercise 5.10

1. For each of the following combinations of interest rates, compounding frequencies and terms, find the value of $i$ and $n$ that would be used in the compound interest formula.
   a. 8%, quarterly compounding, 10 years
   b. 9%, compounded monthly, 7 years
   c. 15%, compounded semiannually, 15 years
   d. 9%, daily compounding using bankers rule, 8 years

2. Calculate the total amount accumulated if Birr 4,500 is deposited for 7 years at 6% interest compounded monthly.
3. For how long must you leave an initial deposit of Birr 100 in a 12% savings account compounded semi-annually to see it grows to Birr 1200?
4. How much do I need to deposit today into a Certificate of Deposit (CD) paying 13.5% compounded monthly in order to have Birr 100,000 in the account in 3 years?
5. Calculate the principal of an investment that will grow to Birr 600,000.00 after two years compounded quarterly at the annual rate of interest 9%.
6. If Mr. Degalla deposits a sum of money in a bank at 8.4% interest rate per year compounded monthly, then how long will it take to double?

## 5.2.2.3 Effective Interest Rate

**Activity 5.9**

In a group of 5 to 7 students visit any three commercial banks and collect information about annual interest rate, compounding frequencies, fixed time deposit (certificate of deposit) and present the report of your finding to the class.

Whether managing a business or our own personal finances, we are often presented with some choice in options for loans, deposits, and other investments. As borrowers, you want to pay as little interest as possible, and so want to be able to find a low interest rate; as lenders, the shoe is on the other foot and you want to seek out a high interest rate. Assuming all other things like, customer handling, branch locations remain the same, you want to find the best rate you can.

In view of the various ways one can compute interest, it is useful to have a method to compare investment strategies. One such device is the **effective interest rate** $r_e$ defined as the simple interest rate that produces the same yield in one year as compound interest.

For any given interest rate and compounding frequency, there is always an equivalent annually compounded rate.

**Definition 5.3**

The annually compounded rate which produces the same results as a given interest rate and compounding is called the equivalent annual rate (EAR) or the effective interest rate. The original interest rate is called the nominal rate.

Let Birr 100 is compounded daily for one year with interest rate 8.98% the future value is Birr 109.394. The equivalent effective interest rate is calculated as:

$A = 100(1 + r_e)^1 = 109.394,$ an exponent of 1 has nothing to do.

$100(1 + r_e) = 109.394.$ Dividing both sides by 100 we obtain

$$1 + r_e = 1.09394,$$

And subtracting 1 from both sides gives us, $r_e = 0.09394 = 9.39\%$ rounded to two decimal places. This number tells us that 9.39% interest rate compounded annually has the same future value as 8.98 % compounded daily.

**Example 1**

Find the equivalent annual rate for 9.5% compounded monthly.

## Solution

Following the procedure given above:

$A = 100(1 + \frac{0.095}{12})^{12} = 109.92.$ Suppose $r_e$ is the rate compounded annually and gives a future value of Birr 109.92.

$$109.92 = 100(1 + R)^1 = 100(1 + r_e)$$

Dividing by 100 both sides, you get $1 + r_e = 1.0992.$ Subtracting 1 from both sides we have $r_e = 0.0992 = 9.92\%.$

From the above examples, we summarize how to find effective interest rate. Thus, to find the effective rate for a given nominal rate and compounding frequency, simply find the future value of Birr 100 in 1 year using the nominal rate and compounding. The effective interest rate (rounded to two decimal places) will be the same number as the amount of interest earned.

Alternatively, if interest is compounded $m$ times a year, then the effective rate must satisfy the equation,

$$P(1+\frac{r}{m})^m = P(1+r_e) \text{ and so } \quad r_e = (1+\frac{r}{m})^m - 1 \qquad 5.5$$

## Example 2

Which of these two banks is offering the best CD rate?

| Bank | Rate | Compounding |
|---|---|---|
| Z | 11.5% | Daily |
| D | 11.5% | Monthly |

### Solution

Both rates are the same, but Z compounds interest daily. Since more frequent compounding means more interest overall, we know that Z will end up paying more interest, and so their rate is the better one.

## Example 3

Which of these two banks is offering the best CD rate?

| Bank | Rate | Compounding |
|---|---|---|
| B | 8.98% | Daily |
| Credit and Saving Associations | 9.05% | Annually |

## Solution

On this example, on the one hand, Workers Credit and Saving Associations rate is higher. Yet on the other, B Bank is offering daily compounding. It is possible that B Bank daily compounding will more than make up for its lower rate, and that despite initial appearances B in the end will be offering more interest. Yet it is also possible that B daily compounding is not enough to catch up with Credit and Saving Association higher rate.

You can resolve the question by finding the effective rate for each case.

For B bank, by using Equation 5.5 , $r_e = (1+\frac{r}{m})^m - 1$

$$r_e = \left(1+\frac{0.0898}{365}\right)^{365} - 1 = 1.09394 - 1 = 0.09394.$$

Hence, the effective interest rate of the B bank is 9.394% . Comparing with the credit and saving association rate 9.05%. The rate given by B bank daily compounding wins by a small margin with lower rate.

Alternatively, suppose, $P = Birr\ 100, t = 1\ year$, we treat separately on compound interest formula as follows: in B Bank at the end of the year Birr 100 matured to Birr 109.39 rounded to two decimal place.

$$A = 100(1+\frac{0.0898}{360})^{360} = 109.394.$$

But for Credit and saving associations, Birr 100 matured to Birr 109.05.

$$A = 100(1+0.0905) = 100(1.0905) = 109.05.$$

From this, we can see that B Bank's daily compounding wins by a small margin with lower rate.

## Exercise 5.11

1. Find the effective rate equivalent to 12.5%, compounded semi-annually.
2. Hagos plans on opening a new saving account. Which of the following is offering the highest rate?

| Bank | Rate | compounding |
|---|---|---|
| Bank A | 2.44 | Annually |
| Bank B | 2.44 | Semi-annually |
| Bank C | 2.44 | monthly |

3. Which of one of the three is offering the best CD rate?

| Institute | Rate | Compounding |
|---|---|---|
| Bank B | 8.98% | monthly |
| Credit and Saving Associations | 9.05% | quarterly |
| XY Microfinance | 9.1% | Semi-annually |

4. Dechase has a life insurance policy with N insurance. The company credits interest to his policy's cash value and offers Dechase the choice of two different options:

| Option | Rate | Compounding |
|---|---|---|
| Daily dividends | 8.00% | Daily( bankers rule) |
| Annual Advancement | 8.33% | Annually |

Which option would give Dechase the most interest?

5. How much interest would a credit union pay on Birr 475 deposit for 300 days if the effective interest rate was 7%?
6. Find the future value of Birr 372,000 at a 15.25% effective rate for 9 months.

## 5.2.3 Annuity

Suppose you borrow Birr 1,400,500 at 12% compounded monthly for 5 years to buy an apartment. The deal is unlikely to be that you get Birr 1,400,500 up front, do nothing for 5 years, and then repay the entire loan at the end, all in a single lump sum. Such an arrangement would not make sense either to you or to the lender.

Instead, the usual arrangement requires you pay off the loan by making payments every month. A timeline for your house loan would instead look more like this:

This loan's consistent and regular payments are an example of an annuity. Now define new terminology.

**Definition 5.4**

An **annuity** is any collection of equal payments made at regular time intervals.

Annuities are very common in business and personal finance. A number of examples are given below; with a few moments' thought you can probably come up with many others.

Some examples of Annuities:

- Monthly payments on a car loan
- Monthly payments on a condominium loan (mortgage).
- Rent payments (until the rent changes)
- Pension on public organization and private company workers (as long as payments do not change)
- Utility bills (same amount paid each month)
- Regularly scheduled deposits to a saving account

**Activity 5.10**

How is "Iqub" managed in our community? Whether Iqub is annuity or not? Discuss in a group.

In regularly scheduled deposit, the annuity payments are made for the purpose of accumulating a sum of money at a future date. In others, such as a mortgage or car loan, the payments are being made in exchange for a sum of money received at the start of the annuity. In both cases, the sum of money involved is important enough to have a name attached to it.

## Example 1

Distinguish between present value and future value.

a. Tesfaye borrowed Birr 160,000 to buy a house. To pay off this mortgage loan, he agreed to make payments of Birr 1,735.52 per month for 30 years. Since his mortgage payments are all equal, and are made at regular intervals, they constitute an annuity. Was the amount he borrowed this annuity's present value or future value?

b. A teacher deposits Birr 250 from each month salary into savings account at work. He will keep this up for the next 40 years, at which time he plans to retire, hopefully having accumulated a large balance in his account. Since equal payments are being made into the account at regular intervals, this is an annuity. Is the value of the account a present value or a future value when the teacher retires?

### Solution

a. Birr 160,000 was received at the start of the annuity payments. Therefore, it would be the present value of the annuity.

b. The value of the account when the teacher reaches retirement would be the future value.

## Timing of the payment

Our definition of an annuity demands that payments be made at regular intervals (annually, monthly, weekly, daily, etc.), but it is silent about the timing of the payments within each interval. In Example b, the teacher agreed to pay Birr 250 monthly for 40 years. This clearly fits the definition of an annuity. But does the payment begin right away (the payment due at the start of each month) or can he wait 30 days before making the first payment (the payment is due at the end of each month)? There are two types of annuities based on timing of payments.

## Definition 5.5

An **ordinary annuity** is an annuity whose payments are made at the end of each time period. An **annuity due** is an annuity whose payments are made at the beginning of each time period.

## Example 2

Distinguish between ordinary annuities and annuities due.

a. Kong took out a car loan on May 7. Payments will be made monthly. The first payment is not due until June 7 (the second will be due on July 7, etc.). Is this an ordinary annuity or an annuity due?

b. Wintana rents a house in Birr 35,000 per year for the next 6 years. She paid the first Birr 35,000 right away. Her second payment will come a year from now, at the start of the second year, and so on. Is this an ordinary annuity or an annuity due?

### Solution

a. Because his first payment is not made until the end of the first month, the second at the end of the second month, and so on, his car payments are an ordinary annuity.

b. Since the payments come at the start of each year, it is an annuity due.

## Exercise 5.12

1. Each of the following problems describes an annuity. Determine whether the amount indicated is the annuity's present value or future value.

   a. Mr. Worku just attended a sales presentation held by the investment company that administers his employer's savings plan. According to the sales representative, if he puts just Birr 1,250 into the plan each month he could be a millionaire when he hits retirement age. Is the Birr 1,000,000 projection a present or future value?

b. City administration won a judgment against a former contractor that will require the contractor to pay the city Birr 52,000.00 per year for the next 20 years. Instead of having to make payments for the next 20 years, the contractor offers to pay the county Birr 835,000.00 in one lump sum today. Would this be a present value or future value?

c. Fifteen years ago Tadesse quit smoking, and instead of spending Birr 70.00 a week on cigarettes he put that money in a special savings account each week. Is the value of his account after 10 years is a present or future value?

2. Each of the following problems describes an annuity. Determine whether it is an ordinary annuity or an annuity due. Justify your answer.

a. Beshir got a share of a company, which will pay him Birr 100,000.00 every year for 20 years, starting right away.

b. At the end of each quarter, a private bank pays each of its shareholders a dividend of 200 Birr for each share owned.

c. Every month I deposited from my salary to our organization credit and saving cooperative union account Birr 400.00.

## 5.2.3.1 Annuities Computation

At this point, you'll begin to develop the mathematical tools need to work with annuities. At first, you will focus just on finding future values, though later on you address present values.

### A. The future value of an ordinary annuity

Suppose that you deposit Birr 1,000 at the end of each year into a saving account that earns 7.5% compounded annually. Assuming you keep the payments up, how much would your account be worth in 10 years? Your annual payments constitute an ordinary annuity, and since we are asking about their accumulated value with interest at the end, we are looking for the future value.

There are three different approaches to develop annuity formula, the Chronological Approach, the Bucket Approach and the Annuity Factor Approach.

## Approach 1: The Chronological Approach

A natural way to approach this problem is to build up the account value year by year, crediting interest as it comes due and adding new payments as they are made.

At the end of the first year, Birr 1,000 is deposited.

At the end of the second year, Birr 1000(1+0.075) =1000(1.075)=Birr 1075.00.

Total amount at the end of year 2 is 1075+1000=2075 (1000 is the monthly deposit).

You could continue along this way until the end of the 10 years, the results of which can be summarized in the Table 4 below:

**Table 4 Chronological approach**

| Year | Starting Balance | Interest Earned | Deposit | Ending Balance |
|---|---|---|---|---|
| 1 | 0.00 | 0.00 | 1000.00 | 1000.00 |
| 2 | 1000.00 | 75.00 | 1000.00 | 2075.00 |
| 3 | 2075.00 | 155.63 | 1000.00 | 3230.63 |
| 4 | 3230.63 | 242.30 | 1000.00 | 4472.92 |
| 5 | 4472.92 | 335.47 | 1000.00 | 5808.39 |
| 6 | 5808.39 | 435.63 | 1000.00 | 7244.02 |
| 7 | 7244.02 | 543.30 | 1000.00 | 8787.32 |
| 8 | 8787.32 | 659.05 | 1000.00 | 10446.37 |
| 9 | 10446.37 | 783.48 | 1000.00 | 12229.85 |
| 10 | 12229.85 | 917.24 | 1000.00 | 14147.09 |

At the end of year 3, the interest earned, $I = 2075(1 + 0.075) = 2075(1.075) = 155.63$

Ending balance

$= 2075 + 155.63 + 1000$

$= 3230.63$

This chronological approach may be fine in cases where the total number of payments is small, but clearly we have reason to find an alternative.

This approach nicely reflects what goes on in the account as time goes by. But it is obvious that we wouldn't want to use it to find the future value of 30 years of weekly payments.

## Approach 2: The Bucket Approach

Suppose now that you opened up a new account for each year's deposit, instead of making them all to the same account. The first payment of Birr 1,000 is placed in the first bucket. This money is kept on deposit from the end of year 1 until the end of year 10, a total of 9 years. So, at the end of the $10^{th}$ year, its bucket will contain a total of Birr $1{,}000(1.075)^9 = \text{Birr } 1{,}917.24$.

You do the same for each of the other eight payments, and then put them all back together again at the end for the total:

**Table 5. The Bucket approach**

| Payment from | Payment Amount | Years of interest | Future value |
|---|---|---|---|
| 1 | 1,000.00 | 9 | Birr 1,917.24 |
| 2 | 1,000.00 | 8 | 1,783.48 |
| 3 | 1,000.00 | 7 | 1,659.05 |
| 4 | 1,000.00 | 6 | 1,543.30 |
| 5 | 1,000.00 | 5 | 1,435.63 |
| 6 | 1,000.00 | 4 | 1,335.47 |
| 7 | 1,000.00 | 3 | 1,242.30 |
| 8 | 1,000.00 | 2 | 1,155.63 |
| 9 | 1,000.00 | 1 | 1,075.00 |
| 10 | 1,000.00 | 0 | 1,000.00 |
| | Grand total | | Birr 14,147.09 |

For the second payment of Birr 1,000, it is kept on deposit from the end of year 2 until the end of year 10, a total of 8 years. So, at the end of the $10^{th}$ year, its bucket will contain a total of Birr $1{,}000(1.075)^8 = \text{Birr } 1{,}783.4$ . Other payments summarized in Table 5 above.

Notice that we arrived at the same total as with the chronological approach as we would have hoped.

## Approach 3: The Annuity Factor Approach

There is a third alternative. Suppose that instead of payments of Birr 1,000 the payments were Birr 3,000 three times as much. It makes sense that the future value would then be three times as much. So to find the future value of the Birr 3,000.00 annuity, we wouldn't need to start from scratch, we could just multiply the Birr 1,000 annuity's future value by 3 to get

$FV = 3(14{,}147.09) = 42{,}441.27$. In general, a larger or smaller payment changes the future value proportionately. Exploiting this, we can define the future value annuity factor.

### Definition 5.6

For a given interest rate, payment frequency, and number of payments, the future value annuity factor is the future value that would accumulate if each payment were Birr 1.00. We denote this factor with the symbol $S_{n/i}$ where $n$ is the number of payment and $i$ is the interest rate per payment period.

Using Bucket approach, with payment of Birr 1 we get

**Table 6. Bucket Approach with Birr 1 payment**

| Payment from | Payment Amount | Years of interest | Future value |
|---|---|---|---|
| 1 | 1.00 | 9 | 1.91724 |
| 2 | 1.00 | 8 | 1.78348 |
| 3 | 1.00 | 7 | 1.65905 |
| 4 | 1.00 | 6 | 1.54330 |
| 5 | 1.00 | 5 | 1.43563 |
| 6 | 1.00 | 4 | 1.33547 |
| 7 | 1.00 | 3 | 1.2230 |
| 8 | 1.00 | 2 | 1.15563 |
| | 1.00 | 1 | 1.07500 |
| 10 | 1.00 | 0 | 1.00000 |
| | Grand total | | 14.14709 |

Future value of Ordinary annuity

$$S_{\overline{n}|i} = (1 + 0.075)^9 + (1 + 0.075)^8 + \cdots + (1 + 0.075)^1 + (1 + 0.075)^0 = 14.14709$$

Since our payments were actually Birr 1,000 times as much, it is logical that the future value we want is:

$$F.V = 1{,}000(14.14709) = 14{,}147.09.$$

Here, $S_{\overline{i}|n}$ is the Future Value Annuity factor, the formula for future value of ordinary annuity with periodic payment R,

$$FV = R \times S_{\overline{i}|n} \qquad 5.6$$

A formula for $\mathbf{S}_{\overline{\mathbf{i}}|\mathbf{n}}$, future value annuity factor

Let's look back at the bucket approach's calculation of $\boldsymbol{S}_{\overline{i}|n}$. The values that we added up to get $\boldsymbol{S}_{\overline{i}|n}$, were:

$$S_{\overline{i}|n} = (1 + 0.075)^9 + (1 + 0.075)^8 + \cdots + (1 + 0.075)^1 + (1 + 0.075)^0$$

Multiplying this equation both sides by $(1 + 0.075)$ you get

$$(1+0.075)S_{\overline{i}|n} = (1+0.075)^{10} + (1+0.075)^9 + \ldots + (1+0.075)^2 + (1+0.075),$$

$$(1.075)S_{\overline{i}|n} = (1.075)^{10} + (1.075)^9 + \cdots + (1.075)^2 + (1.075).$$

Suppose we add 1 to both sides to get:

$$(1.075)S_{\overline{i}|n} + 1 = (1.075)^{10} + (1.075)^9 + \cdots + (1.075)^2 + (1.075) + 1,$$

the right hand side is rewritten as

$$(1.075)S_{\overline{i}|n} + 1 = (1.075)^{10} + S_{\overline{i}|n}$$

Subtracting $1 + S_{\overline{i}|n}$ from both sides we obtain

$(1.075)S_{\overline{i}|n} - S_{\overline{i}|n} = (1.075)^{10} - 1$ , which leads to

$$S_{\overline{i}|n}(1.075 - 1) = (1.075)^{10} - 1$$

$$S_{\overline{i}|n} = \frac{(1.075)^{10} - 1}{0.075}$$

Evaluating this we get, $S_{\overline{i}|n} = 14.14709$ rounded to 5 digits where $n = 10, i = 0.075 = 7.5\%$. Which is the same annuity factor we calculated earlier.

In general, the future value of annuity factor is

$$S_{\overline{n}|i} = \frac{(1+i)^n - 1}{i} \qquad 5.7$$

Where $i$ represents the interest rate per payment period and $n$ represents the number of payments.

**Activity 5.11**

Derive formula (5.7) by using Geometric series concept from Unit 1.

**Example 1**

How much will I have as a future value if I deposit Birr 5,000 at the end of each year into an account paying 9.5 % compounded annually for 20 years?

## Solution

The payments are equal and at regular intervals, and their timing is at the end of each period, so we have an ordinary annuity. Repeat the above process for any interest rate and number of payments. There was nothing special about $n = 10 \; and \; i = 0.075$. If we have $n = 20 \; and \; i = 0.095$, the same process would work. So the future value of the annuity factor is given as

$$S_{\overline{n}|i} = \frac{(1.095)^{20} - 1}{0.95},$$

the future value will be

$$FV = 5000 S_{\overline{0.095}|20}$$

$$= 5000 \left( \frac{(1+0.095)^{20} - 1}{0.095} \right) = 270{,}611.16.$$

**Example 2**

Suppose that Birr 750 is deposited each year into an account paying 12 % interest compounded annually. What will the future value of the account be for 30years?

## Solution

Since the payments are equal and made at regular intervals, this is an annuity; since the timing of the payments is unspecified, we assume it to be an ordinary annuity. By using formula 5.5 and the annuity factor formula 5.6 for the periodic payment R=750,

$$FV = RS_{\overline{i}n} \quad = 750\left(\frac{(1+0.12)^{30}-1}{0.12}\right)$$

This value is calculated by using calculator as

| Operation | $1 + 0.12 =$ | ^30 | -1 | /12 | *750 |
|---|---|---|---|---|---|
| Result | 1.12 | 29.9600 | 28.9600 | 241.3327 | 180,999.51 |

After 30 years the depositor will have Birr 180,999.51.

## Example 3

Find the future value of quarterly payments of Birr 750 for 5 years, assuming a 12% interest rate.

## Solution

We first determine the values of $i$ and $n$. This is done exactly in the same way we did it with compound interest in Section 5.2. Here $r = 0.12,\ m = 4$ since the payment made quarterly

$i = \frac{r}{m} = \frac{0.12}{4} = 0.03$ and $n = m \times t = 4 \times 5 = 20.$

This implies 20 payments of Birr 750 will be made in five years.

Before we proceed, we need to find the annuity factor. To obtain the factor by using the formula, substitute in these values of $n$ and $i$ to get:

$$S_{\overline{i}n} = \left(\frac{(1+i)^n - 1}{i}\right) = \left(\frac{(1+0.03)^{20} - 1}{0.03}\right)$$

The annuity factor

$$S_{\frac{20}{0.03}} = 26.87037.$$

Now that we have the annuity factor, we can complete the calculation. Plugging into our annuity formula (5.5), we get

$$FV = 750 \times S_{\overline{i}|n} = 750 \times 26.87037 = 20{,}152.778.$$

## Exercise 5.13

1. Ato Tesfaye works in XYZ-company earning a monthly salary of Birr 10,460. He is also a member of the credit and saving cooperative union of his company and deposits 25% of his monthly salary at the end of each month at 7% compounded monthly.
   a. Compute Ato Tesfaye accumulated balance by the end of three years.
   b. How much interest has he earned?
2. You are 25 years old and decide to start saving for your retirement. You plan to save Birr 15,000 at the end of each year (so the first deposit will be one year from now), and will make the last deposit when you retire at age 65. Suppose you earn 8% per year on your retirement savings.
   a. How much will you have saved for retirement?
   b. How much will you have saved if you wait until age 35 to start saving (again, with your first deposit at the end of the year)?
   c. Develop a saving schedule by using the bucket approach.
3. Find the future value annuity factor for a monthly annuity, assuming the term is 15 years and the interest rate is 11.1% compounded monthly.
4. At the end of each month, Rekike deposits Birr 250 into a saving account that pays 12% interest compounded monthly. Assuming that she keeps this up, and that the interest rate does not change, how much will her deposits have grown to after twenty years? How much total interest did Rekike earn?
5. Define the three approaches of calculating annuity and discuss the pros and cons of each approach.

6. Calculate the annuity factor for each of the following cases.

   a. If $n = 20, i = 0.025$ b. If $n = 1, i = 0.075$

   c. If $n = 60, i = 0.0025$ d. If $n = 1, i = 0.1$

## B. The Future Value of an Annuity Due

With an annuity due, payments are made at the beginning of each period rather than the end. Each payment is made earlier, so it stands to reason that an annuity due would have a larger future value than an ordinary annuity, since the payments have longer to earn interest.

To see more, let's revisit the 10 year's Birr 1,000 per year annuity with 7.5% interest, this time, though, as an annuity due. The first payment would earn interest from the start of the first year until the end of the tenth year, for a total of ten years. The second payment would earn interest for nine years, the third payment for eight years, and soon. The process summarized in Table 7.

**Table 7. Future value annuity due factor**

| Payment from | Payment Amount | Years of interest | Future value |
|---|---|---|---|
| 1 | 1,000.00 | 10 | 2061.03156 |
| 2 | 1,000.00 | 9 | 1917.23866 |
| 3 | 1,000.00 | 8 | 1783.47783 |
| 4 | 1,000.00 | 7 | 1659.04914 |
| 5 | 1,000.00 | 6 | 1543.30153 |
| 6 | 1,000.00 | 5 | 1435.62933 |
| 7 | 1,000.00 | 4 | 1335.46914 |
| 8 | 1,000.00 | 3 | 1242.29688 |
| 9 | 1,000.00 | 2 | 1155.625 |
| 10 | 1,000.00 | 1 | 1075 |
| | Grand total | | 15,208.11906 |

Notice that this is the same table we used for the ordinary annuity, except that every payment is being credited with 1 additional year of interest. Since crediting 1 year's extra interest is equivalent to multiplying everything by 1.075, we get our future value by multiplying the ordinary annuity's future value by 1.075.

Trying it out, we see that Birr 14,147.09 (1.075) does indeed equal Birr 15,208.11906.

The future value of annuities due is calculated by the formula:

$$FV = R \times S_{\overline{i}}(1+i) \qquad 5.8$$

where $R$ is the periodic payment.

## Example 4

On New Year's Day 2012, Mamo resolved to deposit Birr 2,500 at the start of each year into a retirement savings account. Assuming that he sticks to this resolution, and that his account earns 8.75% compounded annually, how much will he have after 40 years?

### Solution

The payments are equal and made at the start of each year, so this is an annuity due. Thus,

$$FV = R \times S_{\overline{i}}(1+i) = 2500 \times S_{\overline{i}}(1+0.0875)$$

Before going any further, we must find the annuity factor. Using Formula (5.7) you get

$$S_{\overline{i}} = \left(\frac{(1+i)^n - 1}{i}\right) = \left(\frac{(1+0.0875)^{40} - 1}{0.0875}\right) = 316.0346817$$

Returning to the FV formula by multiplying annuity factor, we get:

$FV = 2500 \times 316.0346817 \times (1+0.0875) = 859{,}219.2911.$

So if Mamo does stick with the program, he will have Birr 859,219.30, or nearly Birr 860,000 after 40 years.

## Exercise 5.14

1. Suppose that you deposit Birr 3,500 into a saving account today, and vow to do the same on this date every year. Suppose that your account earns 12.6% compounded monthly and assume there is no drawing from this account. How much will your deposits grow to in thirty years?
2. Abduilkadir has "Iqub". He will pay Birr 1000 at the beginning of each month for the next five years. How much will he get at the end of the term?
3. Suppose that Ato Kifle deposits Birr 1000 into a saving account on behalf of his daughter at the beginning of each month with annual interest rate of 8.4% compounded monthly for the next five years. How much interest will she have after five years? Compare and discuss this result with problem 2.

### 5.2.3.2 Present Values of Annuity

So far, you have considered annuities whose payments and interest build up toward a future value. Now, you will develop the mathematical tools to deal with annuity present values.

***Definition:*** For a given interest rate, payment frequency, and number of payments, the present value annuity factor is the present value of an annuity at this rate, payment frequency, and number of payments if each payment were Birr 1. You denote this factor with the symbol $a_{\overline{n}|i}$ where n is the number of payments and i is the interest rate per payment period.

You will use these present value annuity factors in much the same way as we did future value annuity factors.

Present value of ordinary annuity

$$PV = R \times a_{\overline{n}|i} \qquad 5.9$$

Present value of annuity due

$$PV = R \times a_{\overline{n}|i}(1 + i) \qquad 5.10$$

Formulas for the Present Value of an Annuity

Suppose that we know the payments, term, and interest rate for an annuity, and want to determine its present value. We can find the future value that this annuity could accumulate to by using:

$$FV = RS_{\overline{i}|n}$$

As we've just seen, we can relate this future value to the present value by means of the compound interest formula

$$FV = PV(1+i)^n \qquad 5.11$$

Equating (5.10) and (5.11) we obtain

$$RS_{\overline{i}|n} = PV(1+i)^n$$

Solving for $PV$ we have

$$PV = R\frac{S_{\overline{i}|n}}{(1+i)^n} \qquad 5.12$$

Therefore, the present value annuity factor

$$a_{\overline{i}|n} = \frac{S_{\overline{i}|n}}{(1+i)^n}, \qquad 5.13$$

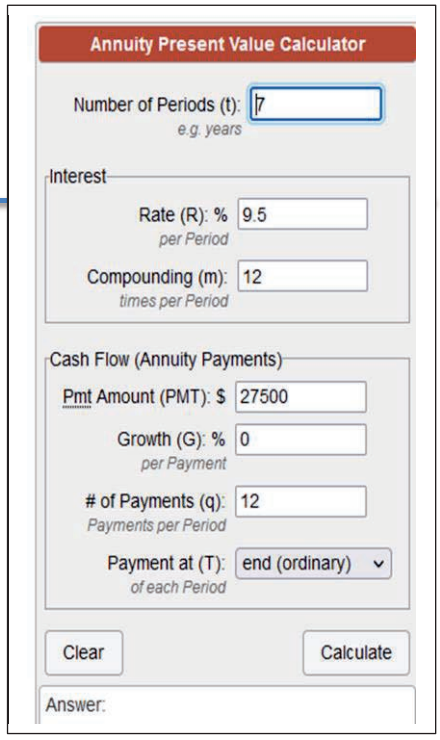
Where $S_{\overline{i}|n}$ is future value annuity factor, $n$ number of periodic payment and $i$ is interest rate of the payment period.

**Example 1**

Find the present value of Birr 27500 per month for seven years at a rate 9.5% compounded monthly.

## Solution

$$i = \frac{0.095}{12} = 0.007917$$



Following our present value formula (5.9) we have:

$PV = R \times a_{\overline{i}n} = 27500 \times a_{\overline{i}n}$

To use Formula (5.13) and by the help of a calculators we obtain

$$S_{\overline{i}n} = \frac{(1+i)^n - 1}{i} = \frac{(1+0.007917)^{84} - 1}{0.007917} = 118.6636$$

$$a_{\overline{i}n} = \frac{S_{\overline{i}n}}{(1+i)^n} = \frac{118.6636}{1.007917} = 61.1838$$

Finally, $PV = 27500 \times 61.1838 = 1{,}682{,}555.93$

The present value $PV =$ Birr 1,682,555.93

https://www.calculatorsoup.com/calculators/financial/present-value-annuity-calculator.php,

## Example 2

Use the annuity factor calculated in Example 1 to find the monthly payment on Birr 800,000 car loan at 9.5% for seven years.

### Solution

On this example the required is the monthly payment R.

$$PV = R \times a_{\overline{i}n}$$

$$800000 = R \times 61.1838$$

Divide both sides by 61.1838

You will get $R = \frac{800000}{61.1838} = 13075.357$. So the monthly payment for the loan Birr 800,000 is around Birr 13,075.

## Exercise 5.15

1. Find the present value annuity factor for a five year annuity with quarterly payments and a 9% interest rate.
2. W/ro Aster borrowed Birr 650,000 for 7 years at 9.5% to buy a house. How much is her monthly payment?
3. Find the present value of Birr 13500 for 10 years at a rate of 12% compounded monthly.

## 5.2.4 Amortization

### Activity 5.12

Assume the family head takes a loan of Birr 10,000 for a year. How does he/she will return the loan?

One of the most familiar applications of Time Value of Money (TVOM) concepts is the amortization schedule. Amortization means "The action or process of gradually writing off the initial cost of an asset." An amortization schedule is a list of balances, payments, and interest charges from the inception of a loan until its payoff.

For example, we have a one-year Birr 1,000 loan at 12% compounded monthly. How do we construct a schedule that shows the allocation of payments to principal and interest over the life of the loan?

The first step is to determine the payment amount that will amortize the loan. This can be accomplished either with the TVOM equation or with spreadsheet. We'll start with PV of an annuity equation

$PV = R\frac{S_{\overline{i}|n}}{(1+i)^n} = Ra_{\overline{i}|n}$, replacing $S_{\overline{i}|n}$ by $\frac{(1+i)^n-1}{i}$ we obtain

$$PV = R\frac{\frac{(1+i)^n-1}{i}}{(1+i)^n} = R\left(\frac{1-(1+i)^{-n}}{i}\right).$$

In this situation the inputs to the **PV** of an annuity equation are as follows,

$$PV = Birr\ 1000.00,\ \ n = 1 \times 12 = 12,\qquad i = 0.12/12 = 0.01$$

And rearranging the above equations to solve R,

$$R = PV\left(\frac{i}{1-(1+i)^{-n}}\right) = 1000 \times \left(\frac{0.01}{1-1.01^{-12}}\right) = 88.85$$

So the payment amount that will amortize the Birr1, 000 loan is Birr 88.85 (rounded). The table that shows series of payments is known as amortization schedule. For Birr 1000 loan with equal monthly payments for one year at 12% the amortization schedule is shown in Table 8 (rounded in two decimal).

**Table 8. Amortization of Birr 1000 with 12% interest rate compounded monthly**

| Month | Principal | Payment | Interest | Payment to principal |
|---|---|---|---|---|
| 1 | 1000.00 | 88.85 | 10.00 | 78.85 |
| 2 | 921.15 | 88.85 | 9.21 | 79.64 |
| 3 | 841.51 | 88.85 | 8.42 | 80.43 |
| 4 | 761.08 | 88.85 | 7.61 | 81.24 |
| 5 | 679.84 | 88.85 | 6.80 | 82.05 |
| 6 | 597.79 | 88.85 | 5.98 | 82.87 |
| 7 | 514.91 | 88.85 | 5.15 | 83.70 |
| 8 | 431.21 | 88.85 | 4.31 | 84.54 |
| 9 | 346.67 | 88.85 | 3.47 | 85.38 |
| 10 | 261.29 | 88.85 | 2.61 | 86.24 |
| 11 | 175.05 | 88.85 | 1.75 | 87.10 |
| 12 | 87.96 | 88.85 | 0.88 | 87.97 |

## Example 1

A family has purchased a home with Birr 30,000 down payment and a Birr 300,000 mortgage. The mortgage will be amortized over thirty years with equal monthly payments. The interest rate on the mortgage will be 12% per year. Determine the monthly payment and compile an amortization schedule.

## Solution

Based on this data, we would like to determine the monthly mortgage payment and compile an amortization table decomposing each of the monthly payments into interest and payment toward principle.

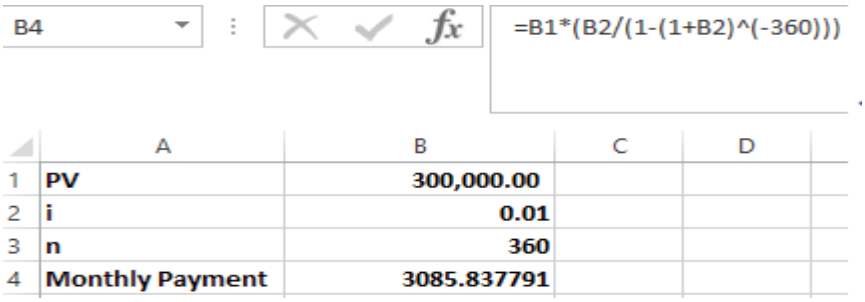
Down payment means a total payment made at the beginning of a payment to secure a loan.

First, we will express annual data as monthly data. Three hundred and sixty $(12 \times 30)$months will elapse before the mortgage is fully paid, and the monthly interest rate will be 1%, or 12% divided by 12. Given this monthly data, monthly mortgage payments are determined as follows:

$$R = PV\left(\frac{i}{1-(1+i)^{-n}}\right) = 300000 \times \left(\frac{0.01}{1-1.01^{-360}}\right) = 3085.838$$

So the family will pay monthly Birr 3085.84 for the next 30 years.

In this particular example, because $n$ is large (360), use of spreadsheet will make computations substantially more efficient.




| | A | B | C | D |
|---|---|---|---|---|
| 1 | PV | 300,000.00 | | |
| 2 | i | 0.01 | | |
| 3 | n | 360 | | |
| 4 | Monthly Payment | 3085.837791 | | |

### Some key points about amortization

Beyond being able to fill in the rows and columns of an amortization table, there are some key points worth noting about amortization.

- The amount of each payment is the same, but the split between interest and principal changes with each payment.
- As the balance is paid down, the portion of each payment dedicated to interest declines. A larger share of the early payments will thus go toward interest than later payments.
- As interest's share of the payments decreases, principal's share of the payments increases. So not only does each payment reduce the amount owed, but also the pace of the reduction is accelerating.

## Exercise 5.16

1. Solomon and Aster took out a thirty year loan for Birr 600,000 at 15%. Calculate their monthly payments. Complete an amortization table for their first 12 monthly payments.
2. Suppose Solomon and Aster took out a fifteen year mortgage instead, for Birr 600,000 and at a 14% rate. Calculate the monthly payments and produce an amortization table for their first 12 monthly payments.
3. Fill in the missing portions of the amortization table. The loan's initial balance was Birr 25000, payments are monthly, and the interest rate is 9%.

| Payment Number | Payment amount | interest amount | Principal | Remaining balance |
|---|---|---|---|---|
| 1 | 500 | | | |
| 2 | 500 | | | |
| 3 | 500 | | | |
| 4 | 500 | | | |
| 5 | 500 | | | |
| 6 | 500 | | | |

a. How many months will it take to amortize the loan?

b. How much total interest will be paid during the loan period?

## 5.2.5 Depreciation

**Activity 5.13**

You bought a mobile phone with Birr 2500. Suppose you want to sell. What is the price of a mobile phone after using, just 1 day, 1 month, and one year? Is there a price change through time? Why?

The cash value of anything that can be owned will change as time goes by. Some things like real estate and collectibles are expected (or at least hoped) to go up in value over time. We call that increase in something's Birr value price appreciation. Other things, though, become less valuable with use and the passing of time. Computers and electronics become obsolete, used cars command lower prices than new ones and business equipment becomes less valuable as it ages and wears out while improved equipment doing the same work comes on the market. The decline in something's cash value is called depreciation.

Let's see some examples on appreciation and depreciation.

**Example 1**

According to a gold expert, the market price for a gram of 24 carat gold is Birr 3650. In an interview in a business media, he states that he expects this particular gold to appreciate at 8% per year for the next 10 years. If his prediction turns out to be correct, what will the price be 10 years from now?

### Solution

This problem is yet another example of a case where even though the 8% appreciation rate is not compound interest, it is a percentage rate of growth and is mathematically equivalent to compound interest. Therefore, we can use the compound interest formula:

$$FV = PV(1+i)^n = 3650 \times (1+0.08)^{10} = 7880.076$$

The price of a gram of gold is expected to more than double in 10 years from now.

## Example 2

Ato Worku just bought a new car for Birr 725,000. According to an online used car pricing service, the value of this car will decline at a 15% annual rate. Assuming this is correct, what will be the car's value over five years?

### Solution

This problem in just the same way as the previous example, except that here the rate is understood to be -15%.

$$FV = PV(1+i)^n = 725000 \times (1-0.15)^5 = 321{,}686.35$$

After five years, the price of that car declined to Birr 321,686.35. The following spreadsheet shows how the price of the car is declining each year in the next five years.

| Year | Starting Price | Decrease | Ending Price |
|---|---|---|---|
| 1 | 725,000 | 108,750 | 616,250.00 |
| 2 | 616,250 | 92,437.5 | 523,812.50 |
| 3 | 523,812.5 | 78,571.88 | 445,240.63 |
| 4 | 445,240.6 | 66,786.09 | 378,454.53 |
| 5 | 378,454.5 | **56,768.18** | 321,686.35 |
| 6 | 321,686.4 | 48,252.95 | 273,433.40 |
| 7 | 273,433.4 | 41,015.01 | 232,418.39 |
| 8 | 232,418.4 | 34,862.76 | 197,555.63 |
| 9 | 197,555.6 | 29,633.34 | 167,922.29 |
| 10 | 167,922.3 | 25,188.34 | 142,733.94 |

As shown in columns 3 and 4, the price is declining at a decelerating rate.

Declining Balance depreciation

In year 5, the cost of the material is 378,454.50. The depreciation amount $d = 378{,}454.50 \times 0.15 = 56{,}768.175.$

At the end of first year the car's price depreciated by Birr 108,750, i.e. 15% of the car's original price. In second year the price reduced by Birr 92,437.5 which was 15% of the new price of the car Birr 616,250.00. When we calculate depreciation in this way, assuming that each year's depreciation is a set percent of a decreasing

value, the amount of annual depreciation decreases with each passing year. For this reason, this is often referred to as declining balance depreciation.

## Exercise 5.17

1. The housing market in Addis Ababa is hot, and a local real estate agent predicted at a meeting of the chamber of commerce that prices would rise by an average of 6.3% per year for the foreseeable future. If the average single-family home costs Birr 1,275,356 today, what would the agent predict in 5 years?
2. In August 2013 the price of an ounce of gold was Birr 640. A commentator on a late-night radio program predicted that the price of gold would rise by an average of 15% per year for the next five years. What price will him projecting in August 2018?
3. Tut bought a boat for Birr 34,750. If the market value declines at a steady 8% annual rate, what will the boat's value be in five years? In ten years?
4. Hayat paid Birr 950,409 for a new truck. She expects that the market value will decline by 11% annually. How much does she expect to be able to sell it for in three years?

## Straight line depreciation

As we have seen in the previous discussion, declining balance depreciation is often a very reasonable way to project the actual market price of something. There is another very commonly used method for calculating depreciation, though. With straight-line depreciation we assume that the price declines by the same Birr amount (not the same percent) each year.

In order to determine the amount of this annual price decline, we must determine the period of time over which we expect the depreciation to occur, this is often termed the item's useful life.

Assume that at the end of its useful life the item still has some salvage value, called its residual value or salvage value.

Once you have determined the useful life and salvage value, we then calculate the total amount of depreciation to be taken and divide this by the useful life. This gives the amount the price will decrease each year. We call this amount the rate of (straight-line) depreciation, or the (straight-line) depreciation rate. Which was expressed as

$$\text{Deperciation} = \frac{\text{Initial Value} - \text{Residual Value}}{\text{Useful life in years}}$$

The value of the item after a period of time calculated from this rate is called the item's depreciated value.

## Example 3

X company purchased a computer for Birr 20,000. The useful life of the computer is 8 years. The computer is assumed to have no salvage value. Find

a) the straight-line depreciation rate,
b) the depreciated value of the computer after 3 years, and
c) the depreciated value after 9 years.

## Solution

a) The computer will lose its full Birr 20,000 initial value in eight years, and since it loses the same amount each year, the depreciation rate is Birr 20,000/8= Birr2500 per year.
b) If the computer's value drops by Birr 2,500 per year for three years, that means it will be worth 20,000-3(2500)=Birr 12,500 at the end of three year.
c) Since the computer's useful life is 8 years, from that time on it has a value of zero. So at nine years, the depreciated value would be Birr 0.

## Example 4

Suppose that a retailer buys a cooler for Birr 24,800. The useful life is 12 years, and the residual value is Birr 1,700. If the cooler is put into use 5 months before the end of the year, determine the depreciated values of the cooler at the end of the year and at the end of next year.

### Solution

First, we determine the annual depreciation amount:

$$\text{Depreciation amount} = \frac{\text{Initial Value} - \text{Residual Value}}{\text{Useful life in years}} = \frac{24800 - 1700}{12}$$

$$= \text{Birr } 1925 \; per \; year.$$

In the first year, the retailer would take 5/12 of a full year's depreciation, or $\left(\frac{5}{12}\right)(1925) = \text{Birr } 802.08$. So at the end of first year, the depreciated value is $24{,}800 - 802.08 = 23{,}997.92$. In next year, a full year's depreciation would be taken, so the end-of-year depreciated value is

$$23{,}997.92 - 1{,}925 = \text{Birr } 22{,}072.92$$

Before leaving this section we compare the two depreciation methods. Both of the types of depreciation that we have considered are useful in different situations. Having seen how each works, it is worthwhile to compare the two methods both mathematically and in how they are most commonly used.

Let's reconsider the case of Example 2, this time with straight-line depreciation. Suppose we assume a useful life of ten years, and assume that at the end of ten years the car has a residual value of Birr 142,733.94.

$$Depreciation \; rate = \frac{725{,}000 - 142{,}733.94}{10} = 58{,}226.61 \; per \; year$$

| Year | Starting Price | Decrease | Ending Price |
|---|---|---|---|
| 1 | 725,000 | 58,226.61 | 666,773.39 |
| 2 | 666,773.39 | 58,226.61 | 608,546.78 |
| 3 | 608,546.78 | 58,226.61 | 550,320.17 |
| 4 | 550,320.17 | 58,226.61 | 492,093.56 |
| 5 | 492,093.56 | 58,226.61 | 433,866.95 |
| 6 | 433,866.95 | 58,226.61 | 375,640.34 |
| 7 | 375,640.34 | 58,226.61 | 317,413.73 |
| 8 | 317,413.73 | 58,226.61 | 259,187.12 |
| 9 | 259,187.12 | 58,226.61 | 200,960.51 |
| 10 | 200,960.51 | 58,226.61 | 142,733.90 |

The slight difference between the 10th year value in the table and the residual value we stated is due to rounding of the straight-line depreciation rate.

Showing this in a table similar to the one in Example 2 with 15% declining balance depreciation, this steady decrease is clearly different from what we saw with percent depreciation.

**Table 9. Comparing percent and straight line**

| Year | Percent | Straight line |
|---|---|---|
| Start | 725,000.00 | 725000.00 |
| 1 | 616,250.00 | 666773.39 |
| 2 | 523,812.50 | 608546.78 |
| 3 | 445,240.63 | 550320.17 |
| 4 | 378,454.53 | 492093.56 |
| 5 | 321,686.35 | 433866.95 |
| 6 | 273,433.40 | 375640.34 |
| 7 | 232,418.39 | 317413.73 |
| 8 | 197,555.63 | 259187.12 |
| 9 | 167,922.29 | 200960.51 |
| 10 | 142,733.94 | 142733.90 |

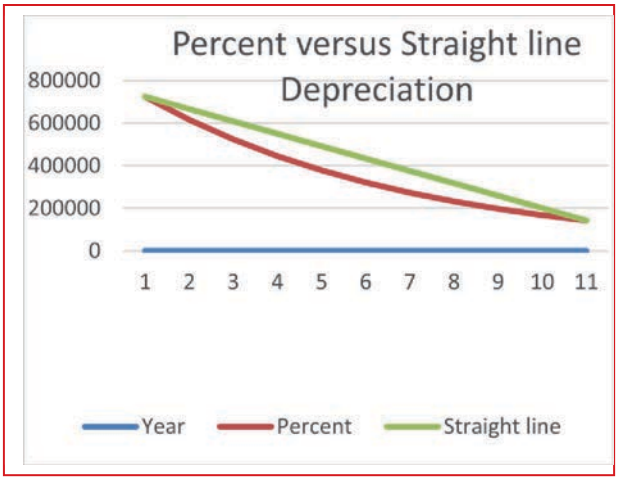



**Figure 5.5 Comparing straight line and Declining balance Method**

You can also illustrate this comparison with a graph (which also illustrates where the name straight-line depreciation comes from), as seen in the Figure above.

While both methods reflect the same reality, which the value of the car is declining with time, there are some significant differences in the values that they predict.

## Exercise 5.18

1. A car that cost Birr 621,450 now is worth Birr 411,050, five years later. What percent depreciation rate would lead to this value?
2. A new refrigerator for a restaurant costs Birr 16,000. The useful life is nine years, and the salvage value is Birr 1,500.
   a. Determine the annual depreciation amount using straight-line depreciation.
   b. Determine the depreciated value of the refrigerator after four years of use.
3. The registrar office bought new filing cabinets for the home office. The total cost was Birr 47,500. The useful life is eight years, and the salvage value is Birr 7,500 . Find the amount of straight-line depreciation per year, and determine the depreciated value of these cabinets after five years of use.

# 5.3 Saving, Investing and Borrowing Money

"Money is the root of all evil"—so the old adage goes. Whether we agree with that sentiment or not, we have to admit that if money is an evil, it is a necessary one. Love it or hate it, money plays a central role in the world and in our lives, both professional and personal. We all have to earn a living and pay bills, and to accomplish our goals, whatever they may be, reality requires us to manage the financing of those goals.

## 5.3.1 Saving

**Activity 5.14**

Discuss why saving is necessary. Which item can we save from the following? Water, natural resource, money or anything what we have.

**What is Money?**

We may define money in terms of functions it performs. Money is something that people use every day. We earn it and spend it but don't often think much about it. Economists define money as any good that is widely accepted as final payment for goods and services. It is defined as "anything which is generally accepted as a medium of exchange in the settlement of all transactions including debt and acts as a measure and store of value".

**Activity 5.15**

Why is Birr said to be money?

There are basically two ways to make money.

1. You work for money. Someone pays you to work for them or you have your own business.

2. Your money works for you. You take your money and you save or invest it.

From the above definitions we derive the following functions of Money:

- Money is a medium of exchange. This means that money is widely accepted as a method of payment. When I go to the market, I am confident that the cashier will accept my payment of money. In fact, Ethiopian paper money carries this statement: "Payable to the bearer on demand" or "**ላምጭው እንዲከፈል ሕግ ያስገድዳል**". This means that the Ethiopian government protects my right to pay with Ethiopian Birr.
- Money is a store of value. If I work today and earn Birr 250, I can hold on to the money before I spend it because it will hold its value until tomorrow, next week,

or even next year. In fact, holding money is a more effective way of storing value than holding other items of value such as corn, which might rot. Although it is an efficient store of value, money is not a perfect store of value. Inflation slowly erodes the purchasing power of money over time.

- Money is a unit of account**.** You can think of money as a yardstick - the device we use to measure value in economic transactions.

Figure 5.6 Currently in use Cash note of FDRE

Once we understand meaning and functions of money, we will see reasons for saving of Money.

## Reasons for saving of Money

There are a variety of reasons to begin saving money. Different people save for different reasons. Here are seven reasons that you may consider for saving your money.

a. Emergency funds
b. A new car, vacations and other luxury items
c. Maximize Interest Rates
d. Retirement
e. Sinking funds
f. A down payment on a house
g. Your education

**Activity 5.16**

Form a group and investigate the following issues: Consider the family of each member in your group and let each student ask his/her family.

1. Whether they save money or not.
2. If yes, why do they save?

After collecting these data allocate each answer under the above reasons for saving.

## Planning a saving program

Sometimes the hardest thing about saving money is just getting started. This step-by-step guide for how to save money can help you develop a simple and realistic strategy, so you can save for all your short- and long-term savings goals.

1. Record your expense;
2. Budget for saving;
3. Find ways you can reduce your spending;
4. Set saving goals;
5. Decide on your priorities;
6. Pick the right tools;
7. Make saving automatics; and
8. Watch your saving grows.

**Activity 5.17**

How public and private companies save for:

a. Retirement,

b. down payment for new house.

c. new car and other fixed assets.

## Savings as investment

**Activity 5.18**

Discuss how saved money is converted to investment.

Is there, any other source of capital to start a new business?

While money doesn't grow on trees, it can grow when you save and invest wisely. Small savings add up to big money.

For instance, if you buy a cup of coffee every day for Birr 5, that adds up to Birr 1825 a year. If you saved that Birr 1825 for just one year, and put it into a savings account or investment that earns 7% a year, it would grow to Birr 2559.66 by the end of five years, and by the end of thirty years, to Birr 13,892.36.

Your "savings" are usually put into the safest places, or products, that allow you access to your money at any time. Savings products include savings accounts, checking accounts, and certificates of deposit.

When you "invest," you have a greater chance of losing your money than when you "save." The money you invest in securities, mutual funds, and other similar investments typically is not federally insured. You could lose your "principal"—the amount you've invested. But you also have the opportunity to earn more money.

**Activity 5.19**

1. Identify kinds of financial institutions.
2. Describe three main factors in choosing a particular institution for saving.

## Saving Institutions

Saving institutions are financial institutions that raise loanable funds by selling deposits to the public. They accept deposits from individuals and firms and use these funds to participate in the debt market, making loans or purchasing other debt instruments such as Treasury bills. Savings institutions include commercial banks, savings and loan associations and credit unions.

### Commercial Banks

Measured by total assets, commercial banks are the most important type of financial institutions. They perform the critical function of facilitating the flow of funds from surplus units to deficit units. In Ethiopia Commercial Banks are either publicly or privately owned.

### Saving and Loan Association

Savings and loans associations (S & Ls) were originally designed as mutual associations, (i.e., owned by depositors) to convert funds from savings accounts into mortgage loans.

Example: Oromia credit and saving, Amhara credit and saving, Dedebit credit and saving, Omo credit and saving.

### Credit unions

Credit unions (CUs) are nonprofit organizations composed of members with a common bond, such as an affiliation with a particular labor union, church, university, or even residential area. Their objective is to serve as an intermediary for those members. For example, different sector employee credit unions.

Because CUs do not issue stock, they are technically owned by the depositors. The deposits are called shares, and interest paid on the deposits is called a dividend. Because CUs are nonprofit organizations, their income is not taxed.

## Exercise 5.19

1. Discuss the main difference between saving institutions.
2. Which of the following account type earn more money to depositors on specific amount and time? Explain your answer in detail.
   a. Saving account. b. Certificate of deposit. c. Checking account.
3. Which of the following account type costs the saving institutes more money on specific amount and time? Explain your answer in detail.
   a. Saving Account b. Certificate of deposit c. Checking account
4. Explain step by step, list of how to save money.

## 5.3.2 Investment

### Activity 5.20

Mr. Tesfaye purchased Great Ethiopians Renassianse Dam (GERD) bond in September 2022 at a cost of Birr 100,000. Was Mr. Tesfye's purchase of GERD bond an investment or not? Discuss it.

An investment is an asset that will eventually provide value that exceeds the initial cost. The term investment can apply to almost any asset, including intangible assets such as education. In terms of the stock market, investing typically refers to the purchase of stocks or bonds. These securities are designed to provide an investor with future value that will exceed their initial cost.

This could be because of:

- A change in market conditions
- A change in the overall supply (the demand for the product is more of the overall supply)
- Because of a direct improvement being made (such as with buying real estate and renovating to increase the value).

Investments can also be made in other assets. Investing in real estate, for example, could mean buying an inexpensive property, renovating to increase its value, and then selling or leasing for more than the original cost.

Investor is a person or organization that puts money into financial schemes, property, etc. with the expectation of achieving a profit or to get advantage.

Some examples of investors are;

a) Commercial banks provide loan to customers with a specific interest rate and return period and buy and sell foreign currency.

b) Real estate developers, different service providers ….

c) A person who learn, a family who teach and nurse the childeren and so on.:

> "The best investment a person can make is in their own ability". Warren Buffet

## 5.3.2.1 Investment goal

**Activity 5.21**

What is your goal in the next five years? Is that an investment goal? Discuss it.

Everybody has investment goals in his/her life, from the old adage of saving for a rainy day to planning a comfortable retirement. Managed funds offer a long-term investment strategy that can help you achieve specific goals. Your choice of fund should reflect the type of life event that you are planning for, as its investment style will determine the returns you can expect over differing timescales.

The three most common types of investment goals are:

- Retirement planning or property purchase over the very long term (15 years or more)
- Life events, such as school fees over the medium term (10-15 years)
- Rainy day or life style funds to finance goals such as to buy car over the medium to shorter term (5-10 years).
- The minimum time horizon for all types of investing should be at least five years.

## 5.3.2.2 Investment Strategy

The term investment strategy refers to a set of principles designed to help an individual investor achieve his/her financial and investment goals. Your investment strategy depends on your personal circumstances, including your age, capital, risk tolerance, and goals. They can vary from conservative (where they follow a low-risk strategy where the focus is on wealth protection) while others are highly aggressive (seeking rapid growth by focusing on capital appreciation).

Investors can use their strategies to formulate their own portfolios or do so through a financial professional. Strategies aren't static, which means they need to be reviewed periodically as circumstances change.

As mentioned above, people can choose to make their investment decisions on their own or by using a financial professional. More experienced investors are able to make decisions and investment choices on their own. Keep in mind that there is no right way to manage a portfolio, but investors should behave rationally by doing their own research using facts and data to back up decisions by attempting to reduce risk and maintain sufficient liquidity.

Because investment strategies depend so heavily on your personal situation and goals, it's important for you to do your research before you commit your capital to any investment.

**Special Considerations.**

Risk is a huge component of an investment strategy. Some individuals have a high tolerance for risk while other investors are risk-averse. Here are a few common risk-related rules:

- Investors should only risk what they can afford to lose,
- Riskier investments carry the potential for higher returns,
- Investments that guarantee the preservation of capital also guarantee a minimal return.

## Example 1

a) A 25-year-old who starts off his/her career and begins saving for retirement may consider riskier investments because he or she has more time to invest and is more tolerant to risk. He/She can also afford to lose some money in the event the market takes a dive because he/she still has time to earn more money. This means he/she can invest in things like stocks and real estate.

b) A 45-year-old, on the other hand, doesn't have a lot of time to put money away for retirement and would be better off with a conservative plan. He or She may consider investing in things like bonds, government securities, and other safe bets.

c) Meanwhile, someone saving for a vacation or home won't have the same strategy as someone saving for retirement. He or She may be better off putting his or her money away in a savings account or a CD for short-term goals like these.

## 5.3.2.3 Types of Investment

There are many types of investments available on the market; from stocks and bonds to mutual funds.

A. **Stock (**Equity**):-** Buying stock is like buying a small fraction of a company; it uses your money to fund the business and you get to enjoy a portion of the profits.

**Ownership of stock** entitles shareholders to four basic rights, unless a specific right is withheld by agreement with the shareholders:

1) Vote. The right to participate in management by voting on matters that come before the shareholders. A shareholder gets one vote for each share of stock owned.

2) Dividends. The right to receive a proportionate part of any dividend. Each share of stock in a particular class receives an equal dividend.

3) Liquidation. The right to receive a proportionate share of any assets remaining after the corporation pays its liabilities in liquidation. Liquidation means to go out of business, sell the assets, pay all liabilities, and distribute any remaining.

4) Preemption. The right to maintain one's proportionate ownership in the corporation. Suppose you own 5% of a corporation's stock. If the corporation issues 100,000 new shares, it must offer you the opportunity to buy 5% (or 5,000) of the new shares. This right, called the preemptive right, is usually withheld from the stockholders.

## Classes of Stock

Corporations issue different types of stock to appeal to a variety of investors. The stock of a corporation may be either common and preferred or Par and no-par.

***Common and Preferred***. Every corporation issues common stock, the basic form of capital stock. Unless designated otherwise, the word *stock* is understood to mean "common stock." Common stockholders have the four basic rights of stock ownership, unless a right is specifically withheld.

***Preferred stock*** gives its owners certain advantages over common stockholders. Preferred stockholders receive dividends before the common stockholders and they also receive assets before the common stockholders if the corporation liquidates. Owners of preferred stock also have the four basic stockholder rights, unless a right is specifically denied.

**Par Value** and **No-Par**. Stock may be par-value stock or no-par stock. Par value is an arbitrary amount assigned by a company to a share of its stock

**No-par stock** does not have par value. But some no-par stock has a **stated value**, which makes it similar to par-value stock.

## Example 1

In Ethiopia most of the companies known as Share Company have been established by selling stocks. In 2020 Dashen bank had a share capital of Birr 3,479,224,000 and Awash Bank had Birr 5,848,271,000. This means Dashen Bank has sold 3,479,224 shares and Awash Bank sold 5,848,271 shares, each of them having a par value of Birr 1000.00.

### Remark

- Authorized stock is the maximum number of shares the company can issue under its present charter.
- Issued stock is the number of shares the company has issued to its stockholders.
- Stock in the hands of a stockholder is said to be outstanding. The total number of shares of stock outstanding at any time represents 100% ownership of the corporation.
- Stockbroker: A stockbroker is a financial professional who executes orders in the market on behalf of clients. A stockbroker may also be known as a registered representative (RR) or an investment advisor.
- A stock exchange, securities exchange, is an exchange where stockbrokers and traders can buy and sell securities, such as shares of stock, bonds, and other financial instruments. Stock exchanges may also provide facilities for the issue and redemption of such securities and instruments and capital events including the payment of income and dividends.

## Example 2

A certain company earned Birr 743,000 in the last quarter, and the company's management has declared a dividend of Birr 450,000. The company has 1,000,000 shares of stock issued. If you own 200 shares of the company's stock, how much will you receive as a dividend?

### Solution

The Birr 450,000 total dividend must be distributed among the shareholders based on the number of shares each one owns; 450,000/1,000,000 shares works out to Birr 0.45 per number of share. Since you own 200 shares, you will receive (200 shares)( 0.45 per number of share) =Birr 90.00.

## Exercise 5.20

1. Explain the main difference between common and preferred shares.
2. List out the basic rights and responsibilities of shareholders.
3. XY Enterprise is a corporation with 3,000 shares. For the current quarter, the company will pay out dividends totaling Birr 37,560.
   a. Find the dividend per share.
   b. If Fantu owns 1,445 shares, how much will she receive?

B. **Bonds/Debt Instruments**: Buying bonds is like having a company takes out a loan from you. It will pay you interest on the principal as well as the full amount later on.

For Example, GERD bond. Here investors are anyone who buy bonds, borrower (issuer of the bond) is Ethiopian Government.

Before proceeding any further, we need to define certain terms that are commonly used with bonds. The issuer of the bond is the entity that creates the bond as a means to borrow funds. Simply, the issuer is the borrower. The par value of a bond is the

amount that will be paid to the bond's owner at maturity. This can also be called the face value of the bond. A bond is said to be redeemed when the borrower pays the bond's owner its par value. For this reason, the par value is also sometimes referred to as the redemption value. Normally, redemption occurs at the bond's maturity date, but when a bond is redeemed early, we say it is called; bonds that may be called are said to be callable.

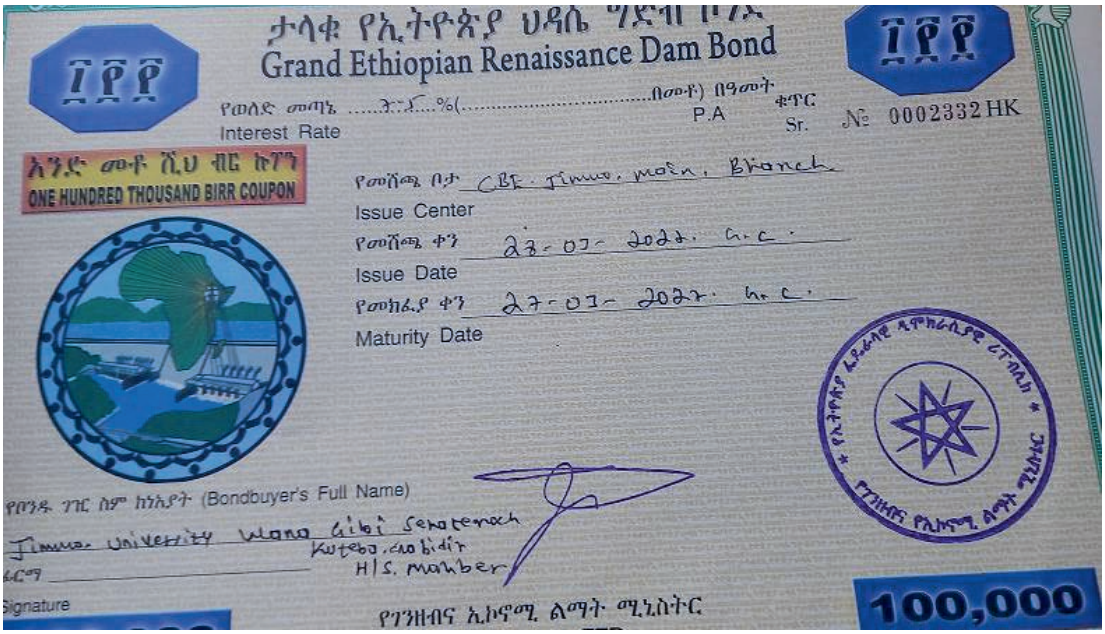



Figure 5.7 GERD Bond

**Characteristics of Bonds**

Most bond certificate have some common basic characteristics including:

- **Face value** is the money amount the bond will be worth at maturity. It is also the reference amount the bond issuer uses when calculating interest payments. For example, say an investor purchases a bond at a premium Birr 1,090 and another investor buys the same bond later when it is trading at a discount for Birr 980. When the bond matures, both investors will receive the Birr 1,000 face value of the bond.
- **The coupon rate** is the rate of interest the bond issuer will pay on the face value of the bond, expressed as a percentage. For example, a 5% coupon rate on Birr

1000 bond means that bondholders will receive $0.05 \times 1000 =$ Birr 50 every year.

- **Coupon dates** are the dates on which the bond issuer will make interest payments. Payments can be made in any interval.
- **The maturity date** is the date on which the bond will mature and the bond issuer will pay the bondholder the face value of the bond.
- **The issue price** is the price at which the bond issuer originally sells the bonds.

**Activity 5.22**

Find an image of a GERD bond and check if all the above characteristics are included on the GERD bond.

**Example 3**

On May 25, 2017, the Ethiopian government issued a Birr 1,000 par value bond with an 8% coupon rate and May 25, 2022, which is maturity date. Interest will be paid semiannually. What payments will the owner of this bond receive?

**Solution**

The owner of this bond will receive the par value, Birr 1,000, on the maturity date of May 25, 2022. Interest payments will be made semiannually, on November 25 and May 25 of each year until maturity. The amount of the interest payments will be

$$I = PRT = 1000 \times 0.08 \times \frac{1}{2} = \text{Birr } 40.$$

The owner of this bond will collect semiannually Birr 40 until maturity date.

### C. Mutual Funds

Mutual funds pool money from several investors and invest it in different asset classes. Each investor sees a return based on performance and the size of their initial contribution. Investors buy shares in mutual funds. Each share represents an investor's part ownership in the fund and the income it generates.

## 5.3.2.4 Return on Investment

**Return on investment**, or ROI, is a metric that evaluates approximately how much value has been gained from an investment relative to the cost. For example, if you had purchased an asset for Birr 10,000.00 and the value appreciates to Birr 12,000.00, then you have gained Birr 2000.00 worth of value for an ROI of 20%.

Return on investment is calculated by the following formula:

$$ROI = \frac{\text{Current Value of investment-cost of investment}}{\text{cost of investment}}$$

### Example 1

You bought an asset currently worth Birr 1000. You initially purchased the asset for Birr 800. The return on investment for that asset would be:

$$ROI = \frac{\text{Current Value of investment-cost of investment}}{\text{cost of investment}} = \frac{1000-800}{800}$$

$$= 25\%$$

### Exercise 5.21

1. Explain the main difference between bond and stock.
2. The coupon rate for a Birr 1,000 par value bond is 7.5%. Find the semiannual interest payment for this bond.
3. A government bond with five years to maturity carries a 6% coupon rate for a Birr 1,000 par value. Find monthly interest payment for this bond. The agent bank calculates interest monthly with 5% annual interest rate on deposit. What payment will receive the owner of the bond at maturity date?
4. Initial investment of Z Company is Birr 2,000,000. After five years the investment grows to Birr 3.5 million. Calculate return on investment of Z Company.

## 5.3.3 Borrowing Money

**Activity 5.23**

Discuss the following in a group.

a. how one finance his/her business;

b. source of finance for business;

c. institutions that finance the business;

d. the advantages and disadvantages of different sources of finance.

Financing is the process of providing funds for business activities, making purchases, or investing. Financial institutions, such as banks, are in the business of providing capital to businesses, consumers, and investors to help them achieve their goals. The use of financing is vital in any economic system, as it allows companies to purchase products out of their immediate reach.

To raise capital for business needs, companies primarily have two types of financing as an option: equity financing and debt financing.

### Equity financing

Equity financing involves selling a portion of a company's equity in return for capital. For example, the owner of company ABC might need to raise capital to fund business expansion. The owner decides to give up 20% of ownership in the company and sell it to an investor in return for capital. That investor now owns 20% of the company and has a voice in all business decisions going forward.

The main advantage of equity financing is that there is no obligation to repay the money acquired through it. Equity financing places no additional financial burden on the company. Since there are no required monthly payments associated with equity financing, the company has more capital available to invest in growing the business.

The drawback of equity financing is:

- The owner has to give investors an ownership percentage of the company
- You have to share your profits with investors
- You give up some control over your company
- It may be more expensive than borrowing.

**Example 1**

Issuing a share is an example of equity financing.

## Debt Financing

Debt financing involves the borrowing of money and paying it back with interest. The most common form of debt financing is a loan. Debt financing sometimes comes with restrictions on the company's activities that may prevent it from taking advantage of opportunities outside the realm of its core business.

The advantages of debt financing are numerous. First, the **lender** has no control over your business. Once you pay the loan back, your relationship with the financier ends. Next, the interest you pay is tax deductible. Finally, it is easy to forecast expenses because loan payments do not fluctuate.

The downside to debt financing is very real to anybody who has debt. Debt is a bet on your future ability to pay back the loan.

**Example 2**

Consider you are an owner of small business company and need Birr 40,000 to finance the business. You can either take out a Birr 40,000 bank loan at a 12% interest rate, or you can sell a 30% stake in your business to your neighbor for Birr 40,000. Which one is a better decision for you?

## Solution

Suppose your business earns a Birr 20,000 profit during the next year. If you took the bank loan, your interest expense (cost of debt financing) would be Birr 4,800, leaving you with Birr 15,200 in profit.

Conversely, had you used equity financing, you would have zero debt (and as a result, no interest expense), but would keep only 70% of your profit (the other 30% being owned by your neighbor). Therefore, your personal profit would only be Birr 14,000, or (70% of Birr 20,000). Therefore, in this scenario taking bank loan is a better decision.

## Source of loan

The main sources of loan or debt financing are saving institutions like commercial banks, saving and loan associations and credit unions. Others include consumer finance companies, insurance companies and private companies.

## Types of loan

Based on the security, a loan may be classified as secured and unsecured loan.

**Secured loan**

With a secured loan, the lender has the right to force the sale of the asset against which the loan is secured if you fail to keep up the repayments. The most common form of secured loan is called a 'further advance' and is made against your home by borrowing extra on your mortgage. Secured loans are mostly suitable for borrowing large amounts of money over a longer term, for example, for home improvements, expansion of a business.

**Unsecured loan**

An unsecured loan means the lender relies on your promise to pay it back. They're taking a bigger risk than with a secured loan, so interest rates for unsecured loans tend to be higher. Unsecured loans are often more expensive and less flexible than secured loans, but suitable if you want a short-term loan.

In accordance with the returning period a loan grouped as short term (1-5 years, middle term (6-10) years and long term loans greater than 10 years.

Other types of loan are discussed as:

**Credit union loan**

Credit unions are mutual financial organizations which are owned and run by their members for their members. Once you've established a record as a reliable saver they will also lend you money but only what they know you can afford to repay.

**Money lines**

Money lines are community development finance institutions that lend and invest in deprived areas and underserved markets that cannot access mainstream finance. They provide money for personal loans, home improvements, back to work loans, working capital, bridging loans, property and equipment purchase, startup capital and business purchase.

**Overdraft**

Overdrafts are like a 'safety net' on your current account; they allow you to borrow up to a certain limit when there's no money in your account and can be useful to cover short term cash flow problems. Overdrafts offer more flexible borrowing than taking out a loan because you can repay them when it suits you, but they're not usually suitable for borrowing large amounts over a long period as the interest rate is generally higher than with a personal loan.

**Buying on credit**

Buying on credit is a form of borrowing. It can include paying for goods or services using credit cards or under some other credit agreement.

## Exercise 5.22

1. A share company needs Birr 20,000,000 to expand the existing company. Then the company looks at the following two sources of finance options:
   a. The company sells 25 % stake by issue 10,000 shares with a value of Birr 1,000 and borrowing the remaining amount from commercial banks.

b. The company borrows all from financial institutions. Assume both options are achievable. At the end of the year the company gets a profit of Birr 5 million. Which option is a better decision for the investor?

2. Write the main advantages and disadvantages of debt financing and equity financing.
3. Investigate company in your local area and explain their source of income to start the business. Categorize the source in to debt and equity financing.
4. Explore sources of loans in your area.
5. When is a loan said to be secured? What is the drawback of unsecured loan?

## 5.4 Taxation

### Activity 5.24

Explain the duties of the government of Ethiopia.

List at least three main duties of the government. Do those duties cost the government? From where does the government get money to cover the cost?

As governments have played a growing role in all economies, they have used increasing amounts of resources for their activities, and taxes have constituted increasing percentages of national income. Either directly or indirectly, the various levels of government provide most education and pay a major proportion of medical bills. They provide national defense, police and fire protection and provide or support a substantial amount of housing, recreation facilities and parklands. They set health standards and ensure adequate water supplies, transportation and other public facilities. They seek to attain a distribution of income regarded as equitable, to stabilize the economy from periods of excessive inflation or unemployment, and to ensure an adequate rate of growth.

According to Richard Musgrave, governmental activities are divided into three parts.

1. **Allocation**: the activities involving the provision of various governmental services to society and thus involving the allocation of resources to the production of these services.
2. **Distribution**: the activities involving in the redistribution of income welfare programs, progressive tax structures and so forth.
3. **Stabilization and growth**: the activities designed to increase economic stability by lessening unemployment and inflation and influencing, if thought desirable, the rate of economic growth.

A tax is a compulsory financial charge or some other type of levy imposed on a taxpayer (an individual or legal entity) by a governmental organization in order to fund government spending and various public expenditures. Taxes consist of direct or indirect taxes and may be paid in money or as its labor equivalent. Taxes are important sources of public revenue.

Ethiopia has a federal tax system, with tax powers and revenues divided between the federal government and the regional states. The power to levy and collect different taxes is allocated either 'exclusively to the federal government; exclusively to the regional states; concurrent to both the federal government and the regional states; [or is] undesignated' (FDRE Constitution, 1995 – Proclamation No. 1/1995).

Concurrent powers are assigned to the federal administration, but the resulting revenue is subject to revenue sharing according to the rules set by the House of the Federation (the higher chamber of parliament). On the other hand, revenues assigned to only one level of government can be fully used at that level, without needing to be shared.

## Objectives of taxation

Governments impose and collect taxes to raise revenue. Revenue generation, however, is not the only objective of taxation, though it is clearly the prime objective. Taxes as a fiscal policy instrument are used to address several other objectives such as:

- Removal of inequalities in income and wealth;
- Ensuring economic stability;
- Changing people's behaviors;
- Beneficial diversion of resources;
- Promoting economic growth.

## Principles of taxation

The compulsory payment by individuals and companies to the state is called taxation. A government imposes taxes to raise revenue to cover the cost of administration, the maintenance of law and order, defense, education, housing, health, pensions, family allowances, etc. In all these, taxes are imposed to provide revenue to cover government expenditure.

**Types of taxes**

Taxes are sometimes referred to as direct or indirect. The meaning of these terms can vary in different contexts, which can sometimes lead to confusion. In economics,

**Direct taxes** refer to those taxes that are paid by the person who earns the income. In direct taxes, the impact and incidence fall on the same person.

By contrast, the cost of **indirect taxes** is borne by someone other than the person responsible for paying them. For example, taxes on goods are often included in the price of the items, so even though the seller sends the payments to the government, the buyer is the real payer. Indirect taxes are sometimes described as hidden taxes because the purchaser of goods or services may not be aware that a proportion of the price is going to the government.

## Exercise 5.23

1. List out major activities of the government.
2. What are the main objectives of taxations in Ethiopia? And explain each objective.
3. Why is a tax said to be direct or indirect taxes?

## 5.4.1 Direct Taxes

According to the Ethiopian tax law direct taxes include all income taxes such as employment income tax, business income tax and land use fee, mining income tax and other income taxes. Generally direct taxes are income based taxes.

### Income tax

Income tax is a very important direct tax. It is an important and most significant source of revenue of the government. "Income" means every sort of economic benefit including gains in cash or in kind, from whatever source derived and in whatever form paid, credited or received.

**"Taxable income"** means the amount of income subject to tax after deduction of all expenses and other deductible items allowed under this Proclamation 286/2002 and Regulations 78/200.

### Sources of Income

Income taxable under this proclamation shall include, but not limited to:

- Income from employment;
- Income from business activities;
- Income derived by an entertainer, musician, or sports person from his personal activities;
- Income from entrepreneurial activities carried on by a non-resident through a permanent establishment in Ethiopia;
- Income from immovable property and appurtenances thereto, income from livestock and inventory in agriculture and forestry, and income from usufruct and other rights deriving from immovable property situated in Ethiopia;
- Dividends distributed by a resident company;
- Profit shares paid by a resident registered partnership;
- Interest paid by the national, a regional or local Government or a resident of Ethiopia, or paid by a non-resident through a permanent establishment that he/she maintains in Ethiopia;

- License fees including lease payments, and royalties paid by a resident or paid by a nonresident through a permanent establishment that he/she maintains in Ethiopia.

The above sources of income are grouped under the following four Schedules:

## Schedules of Income

The proclamation 979/2016 provides for the taxation of income in accordance with four schedules.

- Schedule 'A' Income from employment;
- Schedule 'B' Income from rental buildings;
- Schedule 'C' Income from business;
- Schedule 'D' Income from other sources including royalties' income from technical services rendered outside the country, income from games of chance, dividend income, casual rental of property, interest income and gains from transfer of investment property.

## Schedule 'A': Income from employment

Every person deriving income from employment is liable to pay tax on that income at the rate specified in schedule 'A'. The tax payable on income from employment shall be charged, levied and collected at the following rates. The first Birr 600 (six hundred Birr) of employment income is excluded from taxable income.

**Table 10. Employment income tax rate in different taxable income bracket**

| No. | Taxable Monthly income | Rates of Tax in Percentage |
|---|---|---|
| 1 | Less than Birr 600 | Exemption |
| 2 | [600-1650] on the next Birr 1,050 | 10% |
| 3 | (1650-3200] on the next Birr 1,550 | 15% |
| 4 | (3200-5250]on the next Birr 2,050 | 20% |
| 5 | (5250-7,800] on the next Birr 2,550 | 25% |
| 6 | (7,800-10,900] on the next Birr 3,100 | 30% |
| 7 | Over 10, 900 | 35% |

## Methods of employment income tax computations.

Two methods are used to compute employment income tax.

### 1. Progression method

The amount of tax is calculated for each layer of tax bracket by multiplying the given rate under schedule A For each additional income.

### 2. Deduction methods

$$\textbf{Income Tax} = \textbf{Taxable Income} \times \textbf{tax rate} - \textbf{Deduction}$$

It can be explained as follows:

| No. | Taxable monthly income(Birr) | Rates of tax in percentage | Deduction (Br.) |
|---|---|---|---|
| 1 | < 600 | Exemption | 0 |
| 2 | (600 – 1,650] on the next 1,050 | 10% | 60 |
| 3 | (1,650 – 3,200] on the next 1,550 | 15% | 142.5 |
| 4 | (3,200 – 5,250] on the next 2,050 | 20% | 302.5 |
| 5 | (5,250 – 7,800 ] on the next 2,550 | 25% | 565 |
| 6 | ( 7,800-10,900] on the next ,3100 | 30% | 955 |
| 7 | over 10,900 | 35% | 1,500 |

Deduction is computed by this formula

Deduction = upper taxable income pervious tax bracket tax rate of given bracket cumulative threshold.

Deduction formula illustrated as follow:

$\mathbf{60} = 0.1 \times 600,$

$\mathbf{142.5} = 0.15 \times 600 + 0.05 \times 1050,$

$\mathbf{302.5} = 0.2 \times 600 + 0.1 \times 1050 + 0.05 \times 1550$

$\mathbf{565} = 0.25 \times 600 + 0.15 \times 1050 + 0.1 \times 1550 + 0.05 \times 2050$

### Example 1

Assume Mr. Zelalem earns a monthly salary of $Birr$ 10,460.00. Calculate income tax of Mr. Zelalem.

## Solution

Taxable Income = 10,460.00.

By progression method Birr 10,460 divided into six tax brackets. The summary shown in the following table.

| No. | Amount | Rates of tax in percentage | Taxable amount |
|---|---|---|---|
| 1 | 600 | exempted | 0 |
| 2 | 1050 | 10% | 105 |
| 3 | 1550 | 15% | 232.5 |
| 4 | 2050 | 20% | 410 |
| 5 | 2550 | 25% | 637.5 |
| 6 | 2660 | 30% | 798 |
| Total | 10460 | | 2183 |

Total Income Tax

**Progression Method**

$$2183 = 0 \times 600 + 0.1 \times 1050 + 0.15 \times 1550 + 0.2 \times 2050 + 0.25 \times 2550 + 0.3 \times 2660.$$

**Deduction Method**

10,640 lay in tax bracket (7800-10900) the deduction in this interval is 955 and the rate is 30% as shown in the table

Income tax= $0.3 \times 10460 - 955 = 3138\text{-}955 = 2183$, the result is the same as progression method.

Mr. Zelalem net income after tax is Birr 10,460-2,183=Birr 8,277.00.

## Example 2

Assume Mr. Zelalem got a promotion with salary increment of Birr 2,680.00 on the previous amount of Birr 10,460.00.

a. Calculate the net income of Mr. Zelalem.

b. By what percentage did the net income increase?

c. By what percentage did the income tax increase?

## Solution

a. The new gross salary of Mr. Zelalem is $10460 + 2680 = 13140.00$. Birr 13,140.00 is in the last tax bracket which is 35% taxable rate. By deduction method $income\ tax = 0.35 \times 13140 - 1500 = 3099.$ Net income of Mr. Zelalem is 13140-3099=Birr 10,041.00.(Assume the only deduction is tax)

b. Percentage increase on net income= $\frac{10041-8277}{8277} = 0.213.$ which is 21.3%

c. Percentage increase of the income tax is

Percentage increase on income tax= $\frac{3099-2183}{2183} = 0.4196.$ which is around 41.96% increase on income tax.

## Exercise 5.24

Answer the following questions.

1. Mr. Lodamo earns a basic salary of Birr 14,500. What will be net pay for Mr. Lodamo? Assume there is no deduction other than direct tax.
2. List out all sources of taxable income.
3. What are the main differences between progression and deduction methods?
4. An employee salary increases from Birr 10,000.00 to Birr 21,000.00. What is the percentage change in income tax? What is the percentage change on net income?

### Schedule 'B': Tax on Income from Rental of Buildings

Any income arising from rental of buildings is taxable under schedule 'B'. Rental income includes all form of income from rent of a building and rent of furniture and equipment if the building is fully furnished.

In the lease contract there are two parties involved in renting a building, the lessor and the lessee. The party who grants rent of the building is the lessor**.** The one who leases the property for use is the lessee.

## Taxable Income

a. Gross income includes all payments, either in cash or benefited in kind, received by the lessor.

b. All payments made by the lessee on the behalf of the lessor.

c. The value of any renovation or improvement to the land or the building is also part of taxable income under this schedule if such cost is borne by the lessee in addition to rent payable.

## Deduction

Taxable income from schedule B income is determined by subtracting the allowable deductions from the gross income. Allowable deductions include the following:

**A. For lessors that do not maintain books of accounts**

- taxes paid with respect to the land and buildings being leased; except income taxes; and
- for taxpayers not maintaining books of account, one fifth (1/5) of the gross income received as rent for buildings, furniture and equipment as an allowance for repairs, maintenance and depreciation of such buildings, furniture and equipment.

**B. For lessors that maintain books of accounts**

- For taxpayers maintaining books of account, the expenses incurred in earning, securing, and maintaining rental income, to the extent that the expenses can be proven by the taxpayer and subject to the limitations specified by the Proclamation 979/2016, deductible expenses include (but are not limited to) the cost of lease (rent) of land, repairs, maintenance, and depreciation of buildings, furniture and equipment in accordance with Article 23 of this Proclamation as well as interest on bank loans, insurance premiums. i.e. building 5%, computer and related asset 25%, furniture and equipment 20% and other asset 10% of depreciation base.

### Tax rate

The tax payable on rented houses shall be charged, levied and collected at the following rates:

(a) if the lessors or owners are bodies, they pay thirty percent (30%) of taxable income,

(b) On income of persons according to the Schedule B (hereunder).

**Activity 5.25**

Explain the main difference between allowable deductions mentioned in A and B.

### Schedule B Tax rate and deduction

| No. | Taxable income from rental of building (Income per year) | | Tax rate in percentage | Deduction in Birr |
|---|---|---|---|---|
| | From Birr | To | | |
| 1 | 0 | 7,200 | Exempted | 0 |
| 2 | 7,201 | 19,800 | 10 | 720 |
| 3 | 19,801 | 38,400 | 15 | 1,710 |
| 4 | 38,401 | 63,000 | 20 | 3,630 |
| 5 | 63,001 | 93,600 | 25 | 6,780 |
| 6 | 93,601 | 130,800 | 30 | 11,460 |
| 7 | Over 130,800 | | 35 | 18,000 |

Source: Federal Income Tax Proclamation (No. 979/2016).

**Example 3**

Miss Saba has a building that is available for rent in year 2012. The following are the details of the property let out.

- She has let out for twelve months.
- Actual rent for a month is Birr 30,000
- She paid 15% of the actual rent received as land taxes and 3% as other taxes
- She spent birr 10,000 for maintenance of the building

Other Information in 2012

| Type | Original Cost | Additional cost | Total |
|---|---|---|---|
| Building | 300,000.00 | - | 300,000.00 |
| Equipment | 150,000.00 | - | 150,000.00 |
| Computer and accessories | 10,000/00 | 6000.00 | 16,000.00 |

Compute the taxable income and tax liability

I) He does not maintain any books of accounts in this regard

II) Assumes that Mr. X has maintained books of accounts.

## Solution

I) Annual rental income $12 \times 30{,}000.00 = Birr\ 360{,}000.00$

Allowable deduction

$land\ taxes\ 360{,}000 \times 0.15 = 54{,}000$

$Other\ taxes\ 360{,}000 \times 0.03 = 10{,}800$

$Maintainance\ \left(\frac{1}{5} \times 360{,}000\right) = 72{,}000$

$Total\ Deduction = 136{,}800$

$Taxable\ Income = 360{,}000 - 136{,}800 = 223{,}200$

Then tax liability should be calculated as; Birr 223,200 is in the tax bracket over 130,800 so the rate is 35% and deduction is also Birr 18,000.

Using this numbers

$$Tax\ liability = taxable\ income * tax\ rate - deduction$$

$$= 223{,}200 \times 0.35 - 18{,}000 = Birr\ 60{,}120.00$$

II) For the existence of a book of account

**Depreciation Schedule**

- For building $300{,}000 * 0.05 = 15{,}000.00$
- For Equipment $15{,}000 * 0.2 = 3{,}000.00$
- For computer $16{,}000 * 0.25 = 4{,}000.00$

Annual rental income $Birr\ 360{,}000.00$ is the same for both cases.

Allowable deduction

$$\text{land taxes } 360{,}000 \times 0.15 = 54{,}000.00$$

$$\text{Other taxes } 360{,}000 \times 0.03 = 10{,}800.00$$

$$\text{Maintainance } = 10{,}000.00$$

| | |
|---|---|
| Depreciation expense Building | 15,000.00 |
| Depreciation expense Equipment | 3,000.00 |
| Depreciation expense computer and accessories | 4,000.00 |
| Total Deduction | 96,800.00 |

Taxable Income =Birr 360,000-96,800=Birr 263,200.00

Birr 263,200 is in the tax bracket over 130,800 so the rate is 35% and deduction is also Birr 18,000.

Using this numbers

$$\text{Tax liability} = \text{taxable income} * \text{tax rate} - \text{deduction}$$

$$= 263{,}200 \times 0.35 - 18{,}000 = Birr\ 74{,}120.$$

Liability (generally speaking) is something that is owed to somebody else. Liability can also mean a legal or regulatory risk or obligation.

## SCHEDULE 'C' –INCOME FROM BUSINESS

**Activity 5.26**

What type of business do you know? List out individually and discuss with the classmates.

Business means manufacture or purchase and sale of a commodity with a view to make profit.

The taxable business income of a taxpayer for a tax year shall be determined in accordance with the profit and loss, or income statement, of the taxpayer for the year prepared in accordance with the financial reporting standards, subject to other

provisions of the Proclamation 979/2016, Regulations issued by the Council of Ministers, and Directives issued by the Minister.

### Business Income Tax Rates

1. The rate of business income tax applicable to a body is [30%].
2. The rates of business income tax applicable to an individual are the same as Schedule B tax rate given above.

### Business Category

The Ethiopian tax system classifies businesses into three categories – A, B and C – according to whether the business is incorporated or not, and the size of the business as measured by its turnover. Incorporated taxpayers (corporations) are classified as **Category A** (above 1 million annual turnovers) and face the same tax rate (30%) and administrative requirements regardless of their size. For unincorporated taxpayers, these categories determine the information that firms are required to submit when reporting to the revenue authority.

**Category B**, Businesses are required to submit a profit and loss statement that summarizes the revenues and expenses of the business over the reporting period, but no balance sheet (financial statement) information is required. Their annual turnover is between Birr 500,000 and 1 million.

**Category C**, businesses are not required to keep books of accounts, as firms pay their taxes based on an assessment made by the Ministry of Revenue, MoR. Their annual turnover is below Birr 500,000.

### Example 4

Melat enterprise, unincorporated business has reported earnings before tax of

Birr 80,000 at the tax year ended June 30, 2013. Determine the amount of business income tax.

## Solution

Taxable income Birr 80,000 lies in tax bracket [63,001-93,600] with a rate of 25% and deduction Birr 6,780.00.

Tax payable (business income tax)$= 80{,}000 \times 0.25 - 6{,}780 =$ Birr 13,220.00.

### Schedule 'D' Income: Other Incomes

Other taxable sources of income include royalties, dividends, interest income, winnings from games of chance, and gains on the disposal of investment assets. Hence, any resident who derives income from these sources (as well as any non-resident who derives income from these sources that are attributable to a permanent establishment) is liable for income tax.

| Income source | Applicable tax rate and tax base |
|---|---|
| Royalty | 5% on the gross amount of the royalty |
| Dividend | 10% on the gross amount of the dividend |
| Interest income | 5% of the gross amount of interest in the case of a savings deposit with an Ethiopian financial institution; |
| Game of chance | 15% |
| Gains on disposal of investment asset | 15% for disposal of an immovable asset;<br>30% for disposal of a share or a bond |
| Casual rental income | 15% on the gross amount of rental income |

### Example 5

Ato Tolossa leased his personal car for two months at Birr 36,000 per month. Such income is referred to as casual rental income by the tax expert. How much is the tax paid? Who is liable to pay the tax?

## Solution

Tax on casual rental of property$=0.15 \times 2 \times 36{,}000 = 10{,}800$

The receiver of the income is liable to the tax amount of Birr 10,800.

## Example 6

Workers credit and saving associations have 8, 312 common shares. Each has Birr 1000.00 value of Coop Bank. The bank paid Birr 310 per share in the year ended June 30, 2012 E.C.

a) How much dividend is the association entitled to?

b) How much is the tax to be paid?

c) How much did the association earn on the year?

### Solution

a) Dividend income= Birr $310 \times 8{,}312 =$ Birr $2{,}576{,}720$.

b) According to tax proclamation 979/2016, dividend income is subject to 10% tax rate.

Tax to be paid=$2{,}576{,}720 \times 0.1 =$ Birr $257{,}672.00$.

c) So the association earn Birr $2{,}576{,}720 - 257{,}672.00 =$ Birr $2{,}319{,}048$.

## Exercise 5.25

1. MN Share Company declared to pay 25% dividend to its shareholders. Find the dividend earned and tax to be paid by the following shareholders.
   a. Molla with Birr 300,000 worth of shares.
   b. Jemal with Birr 100,000 worth of shares.
   c. W/ro Gifty with Birr 450,000 worth of shares.
2. If Kelbessa won a lottery worth of Birr 150,000, find the amount of tax he is liable to and his net income.
3. Lapiso rented his loader for 100 days to a contractor at a rate of Birr 15,000 per day. Determine the amount he earns after tax.
4. The top three private commercial banks in Ethiopia gross profit in Birr for 2013 E.C financial year listed below:

| Bank | Profit before tax in Birr | Share capital in Birr |
|---|---|---|
| Awash | 4,823,110,000 | 8,188,948,000 |
| Dashen | 2,426,804,000 | 4,387,996,000 |
| Abyssinia | 2,051,544,000 | 5,182,212,000 |

a. Which tax law governs the banking sector?

b. Calculate the tax paid.

c. Calculate the amount earned by each bank.

d. Calculate earnings per share in percentage and explain it.

5. List any five businesses operated in your area and classify as Category-A, B, and C.

## 5.4.2 Indirect Taxes

Indirect taxes are basically taxes that can be passed on to another entity or individual. They are usually imposed on a manufacturer or supplier who then passes on the tax to the consumer. Many people are not aware of that practically everyone pays taxes, especially indirect taxes. This is because taxes are imposed on almost all the products that we consume. Here are some of the types of indirect taxes.

**Activity 5.27**

Why are many people not aware of paying tax? Discuss in a group.

### Types of indirect taxes

The major types of indirect taxes in Ethiopia are value added tax, turn over tax, excise tax, custom duties and stamp duties.

#### 1. Value added tax (VAT)

This is a sales tax based on the increase in value or price of product at each stage in its manufacture and distribution. The cost of the tax is added to the final price and is eventually paid by the consumer.

The standard rate of VAT in Ethiopia is 15%. However, Ethiopia's tax system also applies differential VAT Zero Rate Tax and some items exempted from VAT.

## Example 1

I purchased a product for resale. Unit price from supplier is Birr 100. 15% VAT is applicable. The selling price in my shop includes 20% markup. What is the final VAT amount paid to the government?

### Solution

I paid the supplier Birr $100 + 0.15 \times 100 =$ Birr 115, the selling price is calculated with 20% markup, i.e. Birr $100 + 0.2 \times 100 = Birr\ 120$, this price is subject to VAT 15%. The sell price is $120 + 0.15 \times 120 =$ Birr 138. VAT liability of the re-seller is Birr 18-15= Birr 3.00.

## Example 2

What is the amount of VAT if the cost of the suit is Birr 7500 (15 % VAT inclusive)?

### Solution

$Birr\ 7{,}500$, includes both cost of the suit and the VAT. Setting $x$ is the cost of the suit.

$Birr\ 7500 = x + 0.15\,x = (1.15)x$ . Dividing both sides by $1.15$ we obtain $x = 7500/1.15 = 6521.74$. Hence, the amount of VAT:

$7500 - 6521.74 =$ Birr 978.26.

### 2. Turnover tax (TOT)

This is an equalization tax imposed on persons not registered for value-added tax to fulfill their obligations and also to enhance fairness in commercial relations and to complete the coverage of the tax system. The standard TOT rate is 2% on goods sold locally and services rendered locally by contractors, grain mills, tractors and combine harvesters, and 10% on other services.

### 3. Excise tax

Excise tax was introduced into the Ethiopian tax system through the Excise Tax Proclamation No. 307/2002. Under this original law, the tax base for the computation of excise taxes differed for locally produced goods and imported goods. For locally produced goods, the tax base was defined as the cost of production. For imported goods, the tax base was defined as the sum of the Cost of product, plus insurance and freight costs (CFI) value and the applicable customs duty.

However, the computation and validation of the tax base for locally produced goods (i.e. the cost of production) created administrative challenges for the tax authority. The new excise tax proclamation (No. 1186/2020) issued in 2020 therefore changed the tax base for locally produced goods to the ex-factory price of the goods, albeit excluding VAT, the cost of excise stamps and the cost of returnable containers.

Excise duty rates now range from 0% to 500% (of either the ex-factory price or CIF value plus customs duties) with the tax being applied to certain demand-inelastic and luxury items, as well as to goods that are assumed to have negative externalities (e.g. fuel, alcohol, tobacco). The tax rates applied to different goods are based on Proclamation 1186/2020.

### 4. Customs duty

Customs duties are levied on goods imported into Ethiopia, are administered and collected by the Customs Commission division of the federal MoR (Ministry of Revenue), and must be paid before the imported items can be released to the importers. Standard tax rates vary between 0% and 35%, with goods often used as raw materials or in capital investment projects typically subject to zero or low rates of duty, and goods generally used as final consumption goods typically subject to high rates of duty.

## 5. Stamp Duty

Stamp duty was introduced in 1998 and is chargeable on various legal, commercial and financial instruments. The legal instrument which regulates stamp duty in Ethiopia is Stamp Duty Proclamation no. 110/1998 and its amendment proclamation no. 612/2008.

### Exercise 5.26

1. What is the amount of VAT if the cost of the suit is Birr 85,100 (15 % VAT inclusive)?
2. Goods are purchased at VAT inclusive price of Birr 166,750. The amount of VAT on this transaction are?
3. A company purchased the following items from a stationery shop.

| Item | Quantity | Unit price before VAT (in Birr) | Total price |
|---|---|---|---|
| Laptop computers | 3 | 36,000.00 | |
| Desktop computers | 5 | 24,000.00 | |
| Hard disk | 2 | 9,000.00 | |
| Mathematics reference book | 10 | 180.00 | |

   i) Complete the table.
   ii) Calculate the total VAT paid.
   iii) What is the total price of the items including VAT?
   iv) If the company wants to pay for the stationery after subtracting a 2% withholding tax before VAT;
      a. What is the amount that will be subtracted for withholding?
      b. What is the amount that the company has to pay for the stationary?
4. A shoe dealer purchased shoes from a shoe company Birr 8,000 worth including VAT. If the dealer sold the shoes for Birr 12,000, find the amount of VAT liable to the dealer.

5. An artist sold his new song to a production company for Birr 1,350,000. What is the royalty tax that should be paid by the artist?
6. Why some listed items exempted from VAT?
7. What do we mean when we talk about VAT at a rate of zero percent?
8. What is the difference between zero percent VAT rate and VAT exempted?
9. List out instruments chargeable with stump duty by referring Proclamation no. 110/1998.

# Summary

A **ratio** is a comparison of two numbers by division.

The ratio of $a$ to $b$ can be expressed as $a: b$ or $a \div b$.

A **rate**, like a ratio, is a comparison of two quantities, but the quantities may have different units of measure.

A rate that has a denominator of 1 is called a **unit rate**.

A rate that identifies the cost of an item per unit is called the unit price.

The rate of change of a given quantity given by the relation:

$$\text{Rate of Change} = \frac{\text{amount of change}}{\text{original amount}} = \frac{\text{final amount} - \text{original amount}}{\text{original amount}}$$

A **proportion** is an equation that states that two ratios are equal.

In the proportion, $a : b = c : d$ with $b \neq 0$ and $a \neq 0$, the four numbers are referred as the terms of the proportions. The first and the last terms $a$ and $d$ are called the extremes; the second and third terms $b$ and $c$ are called the means.

For three quantities $a, b$ and $c$ such that $\frac{a}{b} = \frac{b}{c}$ which is equivalent to $b^2 = ac$ is called the mean proportional between $a$ and $c$.

A **compound proportion** is a situation in which one variable quantity depends on two or more other variable quantities.

Two quantities $x$ and $y$ are said to be an inverse proportion if an increase in $x$ causes a proportional decrease in $y$ and vice-versa.

If $x$ and $y$ are in inverse proportion (vary inversely), $xy = k$ where $k$ is a positive number.

If $x$ and $y$ are in direct proportion (vary directly), $y = kx$ where $k$ is a positive number.

A **percentage** is the numerator of a fraction whose denominator is 100.

The difference between a product selling price and its cost is called markup.

$$\text{Markup} = \text{selling price} - \text{purchase price}$$

**Interest** is what a borrower pays a lender for the temporary use of the lender's money. Or, in other words: Interest is the "rent" that a borrower pays a lender to use the lender's money. The capital originally invested is called the principal. The principal of a loan is the amount borrowed.

The sum of the principal and interest due (or paid) is called the amount (or future value or accumulated value).

A **debtor** is someone who owes someone else money. A **creditor** is someone to whom money is owed.

The amount of time for which a loan is made is called its term.

The future value of a simple interest: $A = P\ (1 + r\ t)$ where $A$ is the future value, $P$ is the principal, $r$ *i*s the simple interest rate per year, and $t$ is the time in years.

Future value or ending amount $A$ is calculated for compound interest by the formula $A = P(1 + r)^n$, where $P$ is the principal, $r$ is the simple interest rate per year and $n$ is the number of years or compounding periods.

Future value of a compound interest $F.V = P(1 + i)^{mt}$

where $F.V$ is amount or future value, $P$ is principal or present value, $i = \frac{r}{m}$, $r$ is annual or nominal rate, $t$ is time in years, and $m$ is the number of conversion periods per year.

**Equivalent annual rate(EAR) or the effective interest rate**: is annually compounded rate which produces the same results as a given interest rate and compounding. The original interest rate is called the nominal rate.

**An annuity** is any collection of equal payments made at regular time intervals.

**An ordinary annuity** is an annuity whose payments are made at the end of each time period.

**An annuity due** is an annuity whose payment are made at the beginning of each time period.

**Amortization** means "The action or process of gradually writing off the initial cost of an asset."

The decline in something's cash value is called *depreciation.*

The increase in something's cash value is called *appreciation.*

A tax is a compulsory financial charge or some other type of levy imposed on a taxpayer (an individual or legal entity) by a governmental organization in order to fund government spending and various public expenditures.

**Usufruct** is the right to enjoy the use and advantages of another's property short of the destruction or waste of its substance.

## Review Exercise

1. What is the ratio of 1.8 km to 900 meters?
2. In a family there are three daughters and a son. What is the ratio of the number of;
   a. Females to the number of people in the family?
   b. Males to the number of females in the family?
3. Allocate a profit of Birr 21,300 of a company among three partners in the ratio of their share of the company1: 2: 3.
4. A group of 15 workers can accomplish a job in 28 days. At the same rate by how many workers can the work be accomplished in 8 days less time?
5. What percent of Birr 52 is Birr 3.12?
6. 8.35% of what amount is Birr 18.37?
7. A 6% tax on a pair of shoes amounts to Birr 102. What is the cost of the pair of shoes?
8. If the average daily wage of a laborer increased from Birr 16.00 to Birr 21.64 in the last three years, what is the rate of increase?
9. A radio recorder sold for Birr 210 has a markup of 25% on the selling price. What is the cost?
10. Ato Alula deposited Birr 3,000 in a saving account that pays 6% interest rate per year, compounded quarterly. What is the amount of interest obtained at the end of seven years? (No deposit or withdrawal was made in these seven years).

11. If you receive 6% interest compounded monthly, about how many years will it take for a deposit at time-0 to triple?
12. Ato Alemu makes regular deposits of Birr 230 at the end of each month for three years. What is the future value of his deposit, if interest rate per year is 9% compounded monthly? What is the amount of interest?
13. At the end of each month Ato Mohammed deposits 10% of his salary in a saving institution that pays annual interest rate of 6% for one year and then 15% for the next three years. If the salary of Ato Mohammed is Birr 1800, find the future value of his deposits at the end of the 4 years.
14. Find the future value annuity factor for a monthly annuity, assuming the term is fifteen years and the interest rate is 7.5% compounded monthly.
15. Find the future value of Birr 857.35 per year for 20 years at 10.5% annual interest rate.
16. A piece of machinery costs Birr 50,000 with an estimated residual value of Birr 7,000 and a useful life of 8 years. It was placed in service on July 1 of the current fiscal year. Determine the accumulated depreciation and book value at the end of the following fiscal year using:
    i) the straight line method.
    ii) the percentage method.
17. A share company needs Birr 10,000,000 to expand the existing company. From the following which option gives better economic advantage for the company?
    a. A share issuing 5,000 shares with a value of Birr 1,000 and borrowing the remaining amount from commercial banks.
    b. Issuing 10,000 shares with par value Birr 1,000.
    c. Borrowing Birr 10,000,000 from commercial banks.

# TABLE OF RANDOM SAMPLING

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 11164 | 36318 | 75061 | 37674 | 26320 | 75100 | 10431 | 20418 | 19228 | 91792 |
| 21215 | 91791 | 76831 | 58678 | 87054 | 31687 | 93205 | 43685 | 19732 | 08468 |
| 10438 | 44482 | 66558 | 37649 | 08882 | 90870 | 12462 | 41810 | 01806 | 02977 |
| 36792 | 26236 | 33266 | 66583 | 60881 | 97395 | 20461 | 36742 | 02852 | 50564 |
| 73944 | 04773 | 12032 | 51414 | 82384 | 38370 | 00249 | 80709 | 72605 | 67497 |
| 49563 | 12872 | 14063 | 93104 | 78483 | 72717 | 68714 | 18048 | 25005 | 04151 |
| 64208 | 48237 | 41701 | 73117 | 33242 | 42314 | 83049 | 21933 | 92813 | 04763 |
| 51486 | 72875 | 38605 | 29341 | 80749 | 80151 | 33835 | 52602 | 79147 | 08868 |
| 99756 | 26360 | 64516 | 17971 | 48478 | 09610 | 04638 | 17141 | 09227 | 10606 |
| 71325 | 55217 | 13015 | 72907 | 00431 | 45117 | 33827 | 92873 | 02953 | 85474 |
| 65285 | 97198 | 12138 | 53010 | 94601 | 15838 | 16805 | 61004 | 43516 | 17020 |
| 17264 | 57327 | 38224 | 29301 | 31381 | 38109 | 34976 | 65692 | 98566 | 29550 |
| 95639 | 99754 | 31199 | 92558 | 68368 | 04985 | 51092 | 37780 | 40261 | 14479 |
| 61555 | 76404 | 86210 | 11808 | 12841 | 45147 | 97438 | 60022 | 12645 | 62000 |
| 78137 | 98768 | 04689 | 87130 | 79225 | 08153 | 84967 | 64539 | 79493 | 74917 |
| 62490 | 99215 | 84987 | 28759 | 19177 | 14733 | 24550 | 28067 | 68894 | 38490 |
| 24216 | 63444 | 21283 | 07044 | 92729 | 37284 | 13211 | 37485 | 10415 | 36457 |
| 16975 | 95428 | 33226 | 55903 | 31605 | 43817 | 22250 | 03918 | 46999 | 98501 |
| 59138 | 39542 | 71168 | 57609 | 91510 | 77904 | 74244 | 50940 | 31553 | 62562 |
| 29478 | 59652 | 50414 | 31966 | 87912 | 87154 | 12944 | 49862 | 96566 | 48825 |
| 96155 | 95009 | 27429 | 72918 | 08457 | 78134 | 48407 | 26061 | 58754 | 05326 |
| 29621 | 66583 | 62966 | 12468 | 20245 | 14015 | 04014 | 35713 | 03980 | 03024 |
| 12639 | 75291 | 71020 | 17265 | 41598 | 64074 | 64629 | 63293 | 53307 | 48766 |
| 14544 | 37134 | 54714 | 02401 | 63228 | 26831 | 19386 | 15457 | 17999 | 18306 |
| 83403 | 88827 | 09834 | 11333 | 68431 | 31706 | 26652 | 04711 | 34593 | 22561 |
| 67642 | 05204 | 30697 | 44806 | 96989 | 68403 | 85621 | 45556 | 35434 | 09532 |
| 64041 | 99011 | 14610 | 40273 | 09482 | 62864 | 01573 | 82274 | 81446 | 32477 |
| 17048 | 94523 | 97444 | 59904 | 16936 | 39384 | 97551 | 09620 | 63932 | 03091 |
| 93039 | 89416 | 52795 | 10631 | 09728 | 68202 | 20963 | 02477 | 55494 | 39563 |
| 82244 | 34392 | 96607 | 17220 | 51984 | 10753 | 76272 | 50985 | 97593 | 34320 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 96990 | 55244 | 70693 | 25255 | 40029 | 23289 | 48819 | 07159 | 60172 | 81697 |
| 09119 | 74803 | 97303 | 88701 | 51380 | 73143 | 98251 | 78635 | 27556 | 20712 |
| 57666 | 41204 | 47589 | 78364 | 38266 | 94393 | 70713 | 53388 | 79865 | 92069 |
| 46492 | 61594 | 26729 | 58272 | 81754 | 14648 | 77210 | 12923 | 53712 | 87771 |
| 08433 | 19172 | 08320 | 20839 | 13715 | 10597 | 17234 | 39355 | 74816 | 03363 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 10011 | 75004 | 86054 | 41190 | 10061 | 19660 | 03500 | 68412 | 57812 | 57929 |
| 92420 | 65431 | 16530 | 05547 | 10683 | 88102 | 30176 | 84750 | 10115 | 69220 |
| 35542 | 55865 | 07304 | 47010 | 43233 | 57022 | 52161 | 82976 | 47981 | 46588 |
| 86595 | 26247 | 18552 | 29491 | 33712 | 32285 | 64844 | 69395 | 41387 | 87195 |
| 72115 | 34985 | 58036 | 99137 | 47482 | 06204 | 24138 | 24272 | 16196 | 04393 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 07428 | 58863 | 96023 | 88936 | 51343 | 70958 | 96768 | 74317 | 27176 | 29600 |
| 35379 | 27922 | 28906 | 55013 | 26937 | 48174 | 04197 | 36074 | 65315 | 12537 |
| 10982 | 22807 | 10920 | 26299 | 23593 | 64629 | 57801 | 10437 | 43965 | 15344 |
| 90127 | 33341 | 77806 | 12446 | 15444 | 49244 | 47277 | 11346 | 15884 | 28131 |
| 63002 | 12990 | 23510 | 68774 | 48983 | 20481 | 59815 | 67248 | 17076 | 78910 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 40779 | 86382 | 48454 | 65269 | 91239 | 45989 | 45389 | 54847 | 77919 | 41105 |
| 43216 | 12608 | 18167 | 84631 | 94058 | 82458 | 15139 | 76856 | 86019 | 47928 |
| 96167 | 64375 | 74108 | 93643 | 09204 | 98855 | 59051 | 56492 | 11933 | 64958 |
| 70975 | 62693 | 35684 | 72607 | 23026 | 37004 | 32989 | 24843 | 01128 | 74658 |
| | | | 85812 | 61875 | 23570 | 75754 | | | |